

# TOBYQA: A Time-Augmented Trust-Region Method for Derivative-Free Optimization in Dynamic Environments

Haoyu Yao<sup>1</sup> and Pengcheng Xie<sup>2,†</sup>

<sup>1</sup>School of Computer Science and Technology, Xi'an Jiaotong University, Xi'an, China ,  
julius06@foxmail.com

<sup>2</sup>Lawrence Berkeley National Laboratory, Berkeley, CA, United States , pxie@lbl.gov

<sup>†</sup>Corresponding author.

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## Abstract

Derivative-free optimization (DFO) faces major challenges in dynamic scenarios where the observation channel drifts continuously over time and function evaluations are subject to observation noise. Conventional trust-region methods generally presume a static landscape, leading to stale-data bias, improper radius updates, and an inability to distinguish spatial descent from temporal drift in the observed function values. We propose TOBYQA (Time-augmented Optimization BY Quadratic Approximation), a regularized DFO framework that jointly incorporates spatial geometry and temporal variations within a unified quadratic interpolation system. TOBYQA augments the classical KKT interpolation system with a ridge term on the  $(1, 1)$ -block, which accommodates observation noise and ensures well-posedness under a mild full-column-rank condition on the constraint block. This relaxes the stringent geometric poisedness requirements of classical methods. An explicit linear-in-time drift term is embedded into the interpolation model, which enables a drift-compensated ratio test that subtracts the estimated temporal component from the observed reduction. Also, a diagonal affine-scaling adapts the search geometry to local variation in curvature. Benchmark evaluations on the Moré–Wild 88-function suite across various temporal drifts and dimensions ( $n \in \{6, 10, 20\}$ ) demonstrate that TOBYQA exhibits enhanced robustness under drift, while maintaining competitive performance in static environments.

**Keywords:** Derivative-free optimization; trust-region methods; affine scaling; time-varying optimization.

## 1 Introduction

Consider the problem of tuning the hyperparameters of a time-series forecasting model on financial data. The hyperparameter vector  $\mathbf{x} \in \mathbb{R}^n$  is evaluated by training multiple models on a held-out validation set of market data and measuring their loss. Because financial markets do not repeat, the validation window necessarily rolls forward with physical time. Consequently, the returned validation loss  $F(\mathbf{x}, t)$  depends on the query time  $t$  in addition to  $\mathbf{x}$  itself. Each evaluation entails a full retraining and backtest pass, which strictly limits the total number of queries; furthermore, no mechanism exists to re-query a previously visited point under exactly identical market conditions.

A second instance arises in the data-driven design of functional materials. In recent work on the accelerated design of magnetic high-entropy alloys, Li, Shan, Zhang and Shek train machine-learning surrogates of the saturation magnetization  $M_s$  and Vickers hardness  $H$  as functions of the alloying composition  $\mathbf{x} \in \mathbb{R}^n$ . They search the composition space for Pareto-optimal points through a multi-objective evolutionary algorithm guided by expected-improvement scores derived from the surrogates. Each candidate composition  $\mathbf{x}$  is scored by the current surrogate, but the surrogate itself is periodically re-trained as newly synthesized alloys are added to the training set. Therefore, the score  $F(\mathbf{x}, t)$  observed by the search loop is a dynamic reflection of the underlying physical objective, drifting as the surrogate’s training distribution evolves with physical time. Each re-training pass is expensive, the synthesis budget generating new training data is strictly constrained by experimental cost, and once the model is refreshed, the surrogate at time  $t_1$  cannot be exactly re-queried at  $t_2 > t_1$ . Similar dynamics appear in any iterative design loop where a continuously updated single learned model serves as the scoring function.

Motivated by these examples, we consider a latent static objective  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  that cannot be queried directly. A query at point  $\mathbf{x} \in \mathbb{R}^n$  and time  $t \geq 0$  returns

$$F(\mathbf{x}, t) = \phi_t(f(\mathbf{x})) + \varepsilon(\mathbf{x}, t), \quad \phi_t(\cdot) := \phi(\cdot, t), \quad (1)$$

where  $\phi_t$  is strictly increasing and  $\varepsilon(\mathbf{x}, t)$  is a zero-mean observation error. In the noise-free case, the monotonicity of  $\phi_t$  preserves the minimizers of the latent objective. With observation noise, the minimizer of a single realized function  $F(\cdot, t)$  may be perturbed, but the optimization target remains the minimizer of  $f$ . The time dependence in Equation (1) therefore affects the observation channel rather than the definition of the optimization target.

The model in Equation (1) captures several structural features that an optimization method must accommodate: the latent objective  $f$  lacks an explicit analytic form and is accessible only through pointwise evaluations of the time-varying observation channel; derivative information is unavailable or unreliable, rendering traditional gradient-based optimization methods inapplicable; the returned evaluation value depends on the query time; and the evaluation budget is limited, requiring the method to extract spatial information from a small number of drifting observations.

The derivative-free optimization problem in the static case (where  $F$  depends solely on  $\mathbf{x}$ ) has been extensively studied. Theoretical foundations and representative algorithms are systematically treated in the monographs of Conn, Scheinberg and Vicente, and Audet and Hare. Among model-based methods, Powell’s NEWUOA and BOBYQA maintain a quadratic surrogate of the objective through minimum Frobenius norm interpolation and have become reference implementations for unconstrained and bound-constrained derivative-free optimization (DFO). Recent work has extended this framework to noisy and stochastic environments: Cartis, Fiala, Marteau and Roberts introduced DFO-LS, which replaces the determined interpolation system with an overdetermined least-squares fit, thereby absorbing observation noise into the residuals; Chen, Menickelly and Scheinberg relaxed the  $\Lambda$ -poisedness condition to a probabilistic one and established convergence under random sampling. When smoothness cannot be assumed, direct-search and evolutionary methods provide complementary tools. However, all these methods are built on the assumption of a static observation channel: evaluations of the same location at different times are presumed to differ only by mean-zero noise.

By contrast, within the model-based DFO framework, the time-varying case remains largely unaddressed. In the gradient-based setting, Fazlyab, Paternain, Preciado and Ribeiro and subsequent studies explored time-varying optimization through a prediction-correction framework, which exploits knowledge of the temporal derivative of  $F$  to track a moving KKT point. In the derivative-free domain, research on dynamic environments is primarily concentrated in the evolutionary computation community (see the survey by Yazdani, Cheng, Tan and Branke); in Bayesian optimization, Morales-Enciso and Branke introduced a temporal kernel function into the Gaussian process model to downweight stale observations. Neither path engages with the algebraic core of the model-based trust-region paradigm, and thus neither inherits its

local convergence guarantees, sample efficiency, or the determinism of methods like NEWUOA, BOBYQA, and DFO-LS. To our knowledge, no model-based trust-region DFO method explicitly handles a time-varying observation channel.

Bridging this gap requires overcoming three primary obstacles. The first is stale-data bias. A model-based DFO method maintains an interpolation set accumulated over past evaluations. When the observation channel drifts, function values attached to older samples no longer reflect the current temporal surface. Consequently, a model fit to mixed-time data biases the recovered gradient and Hessian in directions unpredicted by classical interpolation theory. Second, the standard trust-region ratio test  $\rho_k = (F(\mathbf{x}_k) - F(\mathbf{x}_k + \mathbf{s}_k)) / (\text{predicted reduction})$  loses its validity. The numerator mixes the spatial descent with the temporal drift of the channel between the two evaluations. Thus, depending entirely on the sign of the drift, a step producing genuine spatial improvement might be rejected, or a useless step might be accepted. Third, standard remedies for these issues—resampling the current optimal solution, or shrinking the trust region until drift is negligible—contradict the strictly limited evaluation budget. An effective method must isolate the spatial and temporal contributions to the observed reduction directly within the model itself, rather than expending budget on redundant queries.

In this paper, we propose a model-based trust-region DFO method named TOBYQA (Time-augmented Optimization BY Quadratic Approximation), which addresses these three obstacles within a unified algebraic system. We extend the NEWUOA-style quadratic interpolation framework by appending a linear-in-time column to the interpolation system, introducing a drift coefficient  $\gamma$  that is jointly recovered with the spatial gradient and Hessian, and used to compensate the ratio test. Meanwhile, hard interpolation constraints are replaced by ridge regularization, which absorbs observation noise into the residuals and relaxes classical geometric poisedness requirements. When solving the trust-region subproblem, a diagonal affine-scaling is applied to adapt to the local curvature anisotropy. We prove that its recovery operator yields a clean additive decomposition under linear drift and a bounded perturbation under nonlinear drift.

## 2 Time-Augmented Trust-Region Model

Model-based derivative-free trust-region methods maintain a quadratic surrogate  $Q_k$  of the objective and compute a trial step by minimizing  $Q_k$  inside a trust region whose radius reflects the current model accuracy. The theoretical foundations of this class of methods are developed in the monographs of Conn, Gould and Toint [4] and Conn, Scheinberg and Vicente [3]. The representative implementations are Powell’s NEWUOA [1] and BOBYQA [5], which determine  $Q_k$  from  $m = 2n + 1$  interpolation conditions  $Q_k(\mathbf{x}_i) = f(\mathbf{x}_i)$  together with the minimum-Frobenius-norm closure  $\min \|\nabla^2 Q_k - \nabla^2 Q_{k-1}\|_F$  that resolves the remaining degrees of freedom. Two later developments refine this template along directions that are relevant to the present work. Cartis, Fiala, Marteau and Roberts [6] introduced DFO-LS, in which the determined interpolation system is replaced by an overdetermined least-squares fit and the trust-region update is adjusted to account for a user-specified noise level, so that observation noise is absorbed by the residuals rather than forced into the recovered Hessian. Chen, Menickelly and Scheinberg [7] relaxed the deterministic  $\Lambda$ -poisedness condition to a probabilistic one and established convergence when the model is  $\kappa$ -fully linear with sufficiently high probability, permitting the interpolation set to be constructed by random sampling. Closer to the algebraic formulation adopted below, Xie and Yuan [2] considered objectives whose observed values undergo a step-dependent transformation  $T_k$ , and showed that for the class of transformations they identify, the solution shift induced by  $T_k$  leaves the location of the minimizer of the model subproblem invariant across iterations.

None of these methods addresses the regime in which the observation channel itself depends on  $t$ . Time-varying optimization has been studied extensively in the gradient-based setting, where the prediction–correction framework of Fazlyab, Paternain, Preciado and Ribeiro [8] and

its descendants exploit knowledge of the temporal derivative of  $F$  to track a moving KKT point. On the derivative-free side, work on dynamic environments has been pursued primarily in the evolutionary computation community, surveyed by Yazdani, Cheng, Tan and Branke [10], with representative Bayesian-optimization treatments such as that of Morales-Enciso and Branke [9], in which a Gaussian-process model is augmented with a temporal kernel that discounts stale observations through a hyperparameter fitted to the data. These approaches are effective in the regimes for which they are designed, but they do not engage with the algebraic core of the model-based trust-region paradigm and therefore do not inherit the local convergence guarantees, the sample efficiency, or the determinism of the methods reviewed in the preceding paragraph. Conversely, the trust-region methods reviewed there are built on a static observation channel and cannot, without modification, accommodate evaluations that originate from different temporal surfaces of the same underlying objective.

## 2.1 Problem Formulation and Standing Assumptions

We consider the problem of minimizing a static objective  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  that can only be queried through a noisy, time-dependent observation channel. A query at the point  $x \in \mathbb{R}^n$  and time  $t \geq 0$  returns

$$F(x, t) = \phi(f(x), t) + \varepsilon(x, t), \quad (2)$$

where  $\phi(\cdot, t)$  is a strictly monotone transformation in its first argument, so that the minimizer of  $F(\cdot, t)$  coincides with that of  $f$  for every  $t$ , and  $\varepsilon(x, t)$  is a zero-mean observation noise. The optimization target is therefore well defined despite the time dependence of the observation channel.

The algorithm developed below does not access  $\phi$  or  $f$  separately; it sees only the composite values  $F(x_i, t_i)$ . For the purpose of analysis it is convenient to isolate the dependence of  $F$  on  $t$  at a fixed spatial point. We impose three standing assumptions.

**Assumption 2.1** (Smoothness of the objective). *The function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is twice continuously differentiable, with  $L_g$ -Lipschitz gradient and  $L_H$ -Lipschitz Hessian, and is bounded below by  $f_{\text{low}} > -\infty$ .*

**Assumption 2.2** (Temporal regularity). *For every fixed  $x \in \mathbb{R}^n$ , the map  $t \mapsto \phi(f(x), t)$  is continuously differentiable in  $t$  and Lipschitz in  $t$ : there exists  $L_T > 0$  such that*

$$|\phi(f(x), t) - \phi(f(x), s)| \leq L_T |t - s|, \quad \forall x \in \mathbb{R}^n, \forall t, s \geq 0. \quad (3)$$

**Assumption 2.3** (Observation noise). *The noise variables  $\{\varepsilon(x_i, t_i)\}$  are independent, zero-mean, with uniformly bounded variance  $\text{Var}[\varepsilon(x, t)] \leq \sigma^2$  for a known constant  $\sigma \geq 0$ .*

The following bound is immediate and motivates the linear-in-time correction introduced in the next subsection.

**Proposition 2.4** (Temporal sensitivity of the observation channel). *Under Assumptions 2.2 and 2.3, for any  $x \in \mathbb{R}^n$  and any  $t_i, t_j \geq 0$ ,*

$$\mathbb{E}[|F(x, t_i) - F(x, t_j)|] \leq L_T |t_i - t_j| + 2\sigma. \quad (4)$$

*Proof.* By (2) and the triangle inequality,  $|F(x, t_i) - F(x, t_j)| \leq |\phi(f(x), t_i) - \phi(f(x), t_j)| + |\varepsilon(x, t_i)| + |\varepsilon(x, t_j)|$ . The first term is bounded by  $L_T |t_i - t_j|$  by Assumption 2.2. Taking expectations and using  $\mathbb{E}[|\varepsilon|] \leq \sqrt{\text{Var}[\varepsilon]} \leq \sigma$  by Jensen's inequality gives the claim.  $\square$

The bound (4) is first-order in the time separation. At the model-building stage, this motivates the use of a local linear-in- $t$  correction  $\gamma(t - t_k)$  to absorb the dominant component of the temporal variation of  $F$  over the span of the current trust region: the slope  $\gamma$  is the

first-order Taylor coefficient of  $t \mapsto \phi(f(x), t)$  at  $t = t_k$ , and serves within each iteration as the effective drift rate of the observation channel on the current sample set. This linear ansatz is built into the model in the next subsection; the conditions under which the recovered slope  $\gamma$  is consistent with the true rate  $\partial_t \phi(f(x), t)$  are established in Section 4.

## 2.2 The Soft-Constrained Quadratic Model with Drift

Let  $x_k$  denote the trust-region center at iteration  $k$  and  $t_k$  the time at which the corresponding evaluation was performed. Let  $\{(x_i, t_i, F_i)\}_{i=1}^m$  with  $m = 2n + 1$  denote the current interpolation set, and write  $d_i = x_i - x_k$  and  $\theta_i = t_i - t_k$ , so that  $\theta_i \leq 0$  for historical samples. We seek a quadratic-plus-linear-drift model

$$Q_k(x, t) = c + g^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top G (x - x_k) + \gamma (t - t_k), \quad (5)$$

with unknowns  $c \in \mathbb{R}$ ,  $g \in \mathbb{R}^n$ ,  $G = G^\top \in \mathbb{R}^{n \times n}$  and  $\gamma \in \mathbb{R}$ , determined by

$$\min_{c, g, G, \gamma} \frac{1}{2} \|G\|_F^2 + \frac{\rho}{2} \sum_{i=1}^m (Q_k(x_i, t_i) - F_i)^2, \quad (6)$$

where  $\rho > 0$  is a penalty parameter. The formulation (5)–(6) extends the NEWUOA framework of Powell [1] by an explicit linear-in-time term  $\gamma(t - t_k)$  that absorbs the first-order temporal component of the observation channel motivated by Proposition 2.4, and by replacing the hard interpolation conditions  $Q_k(x_i, t_i) = F_i$  with a soft-constrained penalty whose residuals absorb observation noise of magnitude  $\sigma$  in place of forcing it into the recovered Hessian.

It's worthy to note that the classical NEWUOA closure  $\frac{1}{2} \|G - G_{k-1}\|_F^2$  penalizes the change of the model Hessian between successive iterations, on the premise that the interpolation set evolves by a single-point replacement per iteration and the previous model is therefore the natural reference for the new one. Neither premise applies in the present setting: the time-priority replacement rule (24) retires the oldest non-central sample at each iteration, so the interpolation set may change substantially over the span of a few iterations, and the curvature of the observation channel is itself subject to temporal variation through  $\phi$ . The carry-over term would therefore introduce, into the spatial model, a stale-curvature contribution of the same kind that the drift column  $\theta$  is designed to remove. We therefore drop the carry-over and take  $\frac{1}{2} \|G\|_F^2$  as the regularizer, leaving  $G$  to be determined entirely from the current interpolation data; the regularization role on  $G$  is assumed by the  $\frac{1}{\rho} I_m$  term that arises from the soft constraint and appears in the (1, 1)-block of the resulting linear system.

Introducing residuals  $\xi_i := Q_k(x_i, t_i) - F_i$  and multipliers  $\lambda_i$  for the resulting equality constraints, the KKT stationarity conditions with respect to  $G$ ,  $c$  and  $g$  recover the identities of Powell [14]: the model Hessian is given by

$$G = \sum_{i=1}^m \lambda_i d_i d_i^\top, \quad (7)$$

and the multipliers satisfy  $\sum_i \lambda_i = 0$  and  $\sum_i \lambda_i d_i = 0$ . The remaining two stationarity conditions encode the new ingredients of the present formulation: differentiating the Lagrangian with respect to the residual  $\xi_i$  gives  $\rho \xi_i = \lambda_i$ , and differentiating with respect to the drift unknown  $\gamma$  gives

$$\sum_{i=1}^m \lambda_i \theta_i = 0, \quad (8)$$

an additional orthogonality constraint on the multipliers that parallels the spatial conditions  $\sum_i \lambda_i = 0$  and  $\sum_i \lambda_i d_i = 0$ .

Substituting  $\xi_i = \lambda_i/\rho$  into the residual definition and (7) into the quadratic term yields the  $m$  interpolation equations

$$(A\lambda)_i + c + d_i^\top g + \theta_i \gamma + \frac{1}{\rho} \lambda_i = F_i, \quad i = 1, \dots, m, \quad (9)$$

where  $A \in \mathbb{R}^{m \times m}$  has entries  $A_{ij} = \frac{1}{2}(d_i^\top d_j)^2$ . Together with the three orthogonality conditions on  $\lambda$ , the equations (9) form the augmented KKT system

$$\underbrace{\begin{bmatrix} A + \frac{1}{\rho} I_m & \mathbf{1}_m & D & \theta \\ \mathbf{1}_m^\top & 0 & 0 & 0 \\ D^\top & 0 & 0 & 0 \\ \theta^\top & 0 & 0 & 0 \end{bmatrix}}_{=: M} \begin{bmatrix} \lambda \\ c \\ g \\ \gamma \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (10)$$

in which  $D \in \mathbb{R}^{m \times n}$  has rows  $d_i^\top$ ,  $\theta = (\theta_1, \dots, \theta_m)^\top$ ,  $\mathbf{1}_m$  is the all-ones vector, and  $F = (F_1, \dots, F_m)^\top$ . We denote the augmented constraint block by

$$X_{\text{aug}} := \begin{bmatrix} \mathbf{1}_m & D & \theta \end{bmatrix} \in \mathbb{R}^{m \times (n+2)}, \quad (11)$$

whose first  $n+1$  columns coincide with the constraint block of the classical NEWUOA system, and whose last column  $\theta$  is the temporal extension that enables joint recovery of the drift coefficient.

### 2.3 Well-Posedness and the Hard-Constraint Limit

This subsection establishes that the system (10) admits a unique solution under a mild rank condition on the augmented constraint block, and characterizes the behavior of this solution as  $\rho \rightarrow \infty$ .

**Definition 2.5** (Column-rank condition). *The augmented constraint matrix  $X_{\text{aug}}$  of (11) is said to satisfy the column-rank condition if  $\text{rank}(X_{\text{aug}}) = n+2$ .*

**Proposition 2.6** (Nonsingularity). *Let  $\rho > 0$ . If  $X_{\text{aug}}$  satisfies the column-rank condition, then the matrix  $M$  in (10) is nonsingular, and the system admits a unique solution  $(\lambda, c, g, \gamma)$ .*

*Proof.* The  $(1,1)$ -block  $K := A + \frac{1}{\rho} I_m$  is symmetric positive definite, since the entries  $A_{ij} = \frac{1}{2}(d_i^\top d_j)^2$  form a squared Gram matrix and hence  $A \succeq 0$ , while  $\frac{1}{\rho} I_m \succ 0$ . Eliminating  $\lambda$  from the first block row of (10) yields the Schur-complement system

$$X_{\text{aug}}^\top K^{-1} X_{\text{aug}} \begin{bmatrix} c \\ g \\ \gamma \end{bmatrix} = X_{\text{aug}}^\top K^{-1} F.$$

The matrix  $X_{\text{aug}}^\top K^{-1} X_{\text{aug}}$  is the Gram matrix of the columns of  $X_{\text{aug}}$  with respect to the inner product induced by  $K^{-1}$ ; since  $K^{-1} \succ 0$  and  $X_{\text{aug}}$  has full column rank, it is positive definite and hence invertible. The unique  $(c, g, \gamma)$  is recovered from the Schur-complement system, and  $\lambda$  from the first block row as  $\lambda = K^{-1}(F - X_{\text{aug}}[c, g^\top, \gamma]^\top)$ .  $\square$

**Remark 2.7.** *The corresponding nonsingularity result for the unregularized NEWUOA system requires the  $\Lambda$ -poisedness of the sample set [3], which demands that the displacements  $\{d_i\}$  be sufficiently spread within the trust region and is enforced in Powell's implementation through dedicated geometry-improvement steps. The regularization  $\frac{1}{\rho} I_m$  introduced by the soft constraint renders  $K$  strictly positive definite for any sample geometry, so that the nonsingularity of  $M$  reduces to the linear independence of the columns of  $X_{\text{aug}}$ , a strictly weaker requirement. The*

conditioning of  $M$  remains, of course, dependent on the spread of the displacements, since the smallest eigenvalue of the Schur complement  $X_{\text{aug}}^\top K^{-1} X_{\text{aug}}$  scales with  $\sigma_{\min}(X_{\text{aug}})^2$ . What the soft constraint provides is not insensitivity to geometry, but the avoidance of an outright singular system when the geometry temporarily deteriorates, leaving the point-replacement rule free to prioritize criteria beyond geometric poisedness.

**Corollary 2.8** (Axis-probe initialization). *The axis-probe initialization, which places samples at  $x_0$  and at  $x_0 \pm \Delta_0 e_j$  for  $j = 1, \dots, n$  with strictly increasing time stamps  $t_0 < t_1 < \dots < t_{2n}$ , produces an  $X_{\text{aug}}$  of full column rank  $n + 2$ .*

*Proof.* Index the samples so that the center is  $i = 0$  with  $d_0 = 0$ , and the positive and negative probes along axis  $j$  are  $i = 2j - 1$  and  $i = 2j$  with  $d_{2j-1} = +\Delta_0 e_j$ ,  $d_{2j} = -\Delta_0 e_j$ . The submatrix of  $D$  formed by the rows  $2j - 1$ ,  $j = 1, \dots, n$ , equals  $\Delta_0 I_n$ , hence  $\text{rank}(D) \geq n$ ; since  $D \in \mathbb{R}^{m \times n}$ , we conclude  $\text{rank}(D) = n$ . The constant column  $\mathbf{1}_m$  is independent of the columns of  $D$ , for any relation  $\mathbf{1}_m = D\beta$  would imply on row  $i = 0$  that  $1 = d_0^\top \beta = 0$ , which is impossible. It remains to show that  $\theta \notin \text{span}(\mathbf{1}_m, D)$ . Suppose  $\theta = \alpha \mathbf{1}_m + D\beta$  for some  $\alpha \in \mathbb{R}$  and  $\beta \in \mathbb{R}^n$ . Row  $i = 0$  then gives  $\theta_0 = \alpha$ , and rows  $i = 2j - 1$  and  $i = 2j$  give  $\theta_{2j-1} = \alpha + \Delta_0 \beta_j$  and  $\theta_{2j} = \alpha - \Delta_0 \beta_j$ , so that  $\theta_{2j-1} + \theta_{2j} = 2\alpha = 2\theta_0$  for every  $j$ . But the time stamps are strictly increasing, so  $\theta_{2j-1} + \theta_{2j} > 2\theta_0$  for all  $j \geq 1$ , a contradiction.  $\square$

**Proposition 2.9** (Hard-constraint limit). *Fix the data  $\{(x_i, t_i, F_i)\}_{i=1}^m$  and suppose that the column-rank condition holds. As  $\rho \rightarrow \infty$ , the unique solution  $(\lambda^{(\rho)}, c^{(\rho)}, g^{(\rho)}, \gamma^{(\rho)})$  of (10) converges to the unique solution  $(\lambda^*, c^*, g^*, \gamma^*)$  of the hard-constrained augmented system*

$$\begin{bmatrix} A & \mathbf{1}_m & D & \theta \\ \mathbf{1}_m^\top & 0 & 0 & 0 \\ D^\top & 0 & 0 & 0 \\ \theta^\top & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \lambda \\ c \\ g \\ \gamma \end{bmatrix} = \begin{bmatrix} F \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (12)$$

which enforces the strict interpolation conditions  $Q_k(x_i, t_i) = F_i$ . In the noise-free case in which  $f$  is exactly quadratic on the sample set and the observation channel exhibits no temporal drift, the limit satisfies  $c^* = f(x_k)$ ,  $g^* = \nabla f(x_k)$ ,  $\gamma^* = 0$ , and the implicit Hessian  $G^* = \sum_i \lambda_i^* d_i d_i^\top$  coincides with  $\nabla^2 f(x_k)$ .

*Proof.* The stationarity condition  $\rho \xi_i = \lambda_i$  established in Section 2.2 gives  $\xi_i = \lambda_i / \rho$ , and the value of the objective (6) at the optimum is  $\frac{1}{2} \|G^{(\rho)}\|_F^2 + \frac{1}{2\rho} \|\lambda^{(\rho)}\|^2$ . Since this value is bounded above by the objective evaluated at any hard-feasible point, it is bounded uniformly in  $\rho$ . Hence  $\|\lambda^{(\rho)}\|^2 = O(\rho)$  and  $\|\xi^{(\rho)}\| = \|\lambda^{(\rho)}\| / \rho = O(\rho^{-1/2})$ , which tends to zero as  $\rho \rightarrow \infty$ ; the constraint (9) then reduces, in the limit, to the hard interpolation conditions. The hard-constrained system (12) is itself nonsingular under the column-rank condition, by the Schur-complement argument of Proposition 2.6 applied with  $K$  replaced by  $A$  restricted to the column space of  $X_{\text{aug}}$ . Continuity of the solution map in the regularization parameter then yields convergence to the hard-constrained solution. When  $f$  is exactly quadratic on the sample set and the observation is noise-free and drift-free, the right-hand side admits the exact representation  $F_i = f(x_k) + \nabla f(x_k)^\top d_i + \frac{1}{2} d_i^\top \nabla^2 f(x_k) d_i$ , and the stated solution satisfies (12) by direct verification.  $\square$

**Remark 2.10.** *Proposition 2.9 positions the soft-constrained augmented system as a strict generalization of its hard-constrained counterpart: a finite  $\rho$  trades exact interpolation for tolerance to observation noise, while the temporal column is silent on noiseless static data, in which case the model reduces to the unique minimum-Frobenius-norm quadratic interpolant of  $f$  on the sample set. The hard-constrained limit (12) should not be identified with the classical NEWUOA system of Powell [1]: it differs from the latter by the presence of the temporal column  $\theta$  and by the absence of a  $G_{k-1}$  carry-over term in the Frobenius regularizer, for the reasons given in Section 2.2.*

## 2.4 Affine-Scaling Trust-Region Subproblem

The augmented KKT system (10) recovers, in physical coordinates, a spatial gradient  $g$  and an implicit Hessian  $G = \sum_{i=1}^m \lambda_i d_i d_i^\top$ . The trust-region step could in principle be obtained by minimizing the spatial part of the model,  $g^\top s + \frac{1}{2} s^\top G s$ , over the ball  $\|s\|_2 \leq \Delta_k$ . Two considerations argue against the direct formulation. First, the truncated conjugate-gradient method that Powell uses to solve this subproblem [1] has a worst-case iteration count proportional to  $\sqrt{\text{cond}(G)}$ , so on objectives whose curvature varies strongly across coordinate directions this cost dominates the per-iteration work of TOBYQA. Second, a spherical trust region allocates its radius budget inefficiently on an anisotropic landscape: a single radius is imposed on directions of sharp curvature, where it is appropriately conservative, and on directions of flat curvature, where it forces unnecessarily short moves. A diagonal preconditioner of the subproblem addresses both points at negligible cost.

Introduce a diagonal scaling matrix  $S = \text{diag}(s_1, \dots, s_n)$  with  $s_j > 0$ , and define scaled variables by the linear change of coordinates  $s = S\hat{s}$ , equivalently  $\hat{s} = S^{-1}s$ . Substituting into the spatial model gives

$$g^\top s + \frac{1}{2} s^\top G s = \hat{g}^\top \hat{s} + \frac{1}{2} \hat{s}^\top \hat{G} \hat{s}, \quad \hat{g} = Sg, \quad \hat{G} = SGS. \quad (13)$$

We replace the physical spherical constraint  $\|s\|_2 \leq \Delta_k$  with a spherical constraint  $\|\hat{s}\|_2 \leq \Delta_k$  in the scaled coordinates, which corresponds in physical space to the ellipsoidal region  $s^\top S^{-2} s \leq \Delta_k^2$ . The half-axis along coordinate direction  $j$  has length  $s_j \Delta_k$ , so that an appropriate choice of  $s_j$  allocates a longer step budget along directions of small curvature and a shorter budget along directions of large curvature. The scaled subproblem is

$$\min_{\hat{s} \in \mathbb{R}^n} \hat{g}^\top \hat{s} + \frac{1}{2} \hat{s}^\top \hat{G} \hat{s} \quad \text{subject to} \quad \|\hat{s}\|_2 \leq \Delta_k. \quad (14)$$

The physical step is recovered as  $s = S\hat{s}$ .

The scaled Hessian admits the factored form

$$\hat{G} = SGS = \sum_{i=1}^m \lambda_i (Sd_i)(Sd_i)^\top, \quad (15)$$

which is the form actually consumed by the conjugate-gradient solver: Hessian-vector products  $\hat{G}v$  are evaluated as  $\sum_i \lambda_i [(Sd_i)^\top v](Sd_i)$ , requiring neither the explicit assembly of  $\hat{G}$  nor the formation of  $G$  itself. The solver's input is thus the pair  $(\{Sd_i\}_{i=1}^m, \{\lambda_i\}_{i=1}^m)$  together with the scaled gradient  $\hat{g}$ ; no other transformation of the interpolation set is required.

For the choice of  $S$ , we seek to make  $\hat{G}$  approximately isotropic. Diagonalizing the requirement that  $\hat{G}_{jj} = s_j^2 G_{jj}$  be roughly independent of  $j$  leads to the rule

$$s_j = \frac{(|G_{jj}| + \epsilon)^{-1/2}}{\max_{1 \leq l \leq n} (|G_{ll}| + \epsilon)^{-1/2}}, \quad (16)$$

where  $\epsilon > 0$  guards against vanishing diagonal entries. The normalization  $\max_j s_j = 1$  makes  $S$  a contraction, so that the trust region in scaled coordinates is contained in the physical ball of radius  $\Delta_k$ ; no direction is ever assigned a step budget exceeding the nominal radius.

The scaling matrix  $S$  is constructed from the diagonal of the implicit Hessian recovered by the previous iterate's KKT solve, but does not itself enter the KKT system (10). The recovered  $(\lambda, c, g, \gamma, G)$  are determined in physical coordinates and are independent of  $S$ , so that any inaccuracy in  $S$  during early iterations affects only the trust-region step through (14) and leaves the drift estimate  $\gamma$  unaffected.

## 2.5 Drift-Compensated Ratio Test

In the time-varying setting the classical trust-region ratio test cannot be applied directly: the algorithm observes only the noisy, time-dependent surrogate  $F(x, t)$ , so that the actual reduction  $f(x_k) - f(x_k + s)$  that would appear in a static ratio test is not available. The two values that are available,  $F(x_k, t_{\text{old}})$  and  $F(x_k + s, t_{\text{new}})$ , differ in their time arguments and therefore mix the spatial descent the step has achieved with the temporal drift of the observation channel over the interval  $[t_{\text{old}}, t_{\text{new}}]$ . The drift coefficient  $\gamma$  recovered jointly with  $(c, g, G)$  by the KKT system (10) provides a model-based estimate of this temporal component, which the ratio test below uses to compensate.

**Definition 2.11** (Predicted and drift-compensated actual reduction). *Given the model parameters  $(\lambda, c, g, \gamma)$  obtained from (10) and a candidate step  $s$ , the predicted reduction is*

$$\text{Pred}_k := Q_k(x_k, t_k) - Q_k(x_k + s, t_k) = -g^\top s - \frac{1}{2} \sum_{i=1}^m \lambda_i (d_i^\top s)^2. \quad (17)$$

With  $F_{\text{old}} := F(x_k, t_{\text{old}})$  the most recent recorded observation at the current center,  $F_{\text{new}} := F(x_k + s, t_{\text{new}})$  the observation at the trial point, and  $\Delta t_q := t_{\text{new}} - t_{\text{old}}$  the query separation, the drift-compensated actual reduction is

$$\text{Ared}_k := (F_{\text{old}} - F_{\text{new}}) + \gamma \Delta t_q. \quad (18)$$

Both quantities are referred to the same time  $t_k$  on the model side:  $\text{Pred}_k$  evaluates  $Q_k(\cdot, t_k)$  at both arguments, so that the drift term  $\gamma(t - t_k)$  contributes nothing and  $\text{Pred}_k$  measures spatial descent only. On the observation side, the correction  $\gamma \Delta t_q$  in  $\text{Ared}_k$  subtracts the model's estimate of the temporal component of  $F_{\text{old}} - F_{\text{new}}$ , leaving an estimate of the spatial descent that  $\text{Pred}_k$  predicts. The following proposition gives the exact decomposition under additively separable drift, which is the special case of (2) with  $\phi(v, t) = v + \mu(t)$ ; the general monotone channel is treated in Section 4.

**Proposition 2.12** (Compensation error under additive drift). *Suppose the observation channel takes the form*

$$F(x, t) = f(x) + \mu(t) + \varepsilon(x, t), \quad (19)$$

with  $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}$  differentiable and  $\varepsilon$  a zero-mean noise term satisfying Assumption 2.3. Then

$$\text{Ared}_k = (f(x_k) - f(x_k + s)) + [\gamma \Delta t_q - (\mu(t_{\text{new}}) - \mu(t_{\text{old}}))] + (\varepsilon_{\text{old}} - \varepsilon_{\text{new}}), \quad (20)$$

where  $\varepsilon_{\text{old}}, \varepsilon_{\text{new}}$  are the noise realizations at the two evaluations. In particular, if  $\mu$  is linear,  $\mu(t) = \beta t$ , and the recovered slope satisfies  $\gamma = \beta$ , then the middle bracket vanishes and  $\text{Ared}_k = (f(x_k) - f(x_k + s)) + \varepsilon_{\text{old}} - \varepsilon_{\text{new}}$ .

*Proof.* Substituting (19) into (18),

$$\begin{aligned} \text{Ared}_k &= [f(x_k) + \mu(t_{\text{old}}) - f(x_k + s) - \mu(t_{\text{new}})] + \gamma \Delta t_q + \varepsilon_{\text{old}} - \varepsilon_{\text{new}} \\ &= (f(x_k) - f(x_k + s)) + [\gamma \Delta t_q - (\mu(t_{\text{new}}) - \mu(t_{\text{old}}))] + \varepsilon_{\text{old}} - \varepsilon_{\text{new}}, \end{aligned}$$

which is (20). The linear case is immediate.  $\square$

The decomposition (20) isolates three contributions to  $\text{Ared}_k$ . The first is the spatial descent  $f(x_k) - f(x_k + s)$ , the quantity  $\text{Pred}_k$  is designed to estimate. The second is the drift compensation residual, which vanishes when the recovered  $\gamma$  matches the chord slope of  $\mu$  over  $[t_{\text{old}}, t_{\text{new}}]$  and is exactly zero in the linear-drift case. The third is the difference of two noise realizations, of magnitude  $2\sigma$  in expectation under Assumption 2.3. This motivates a two-regime acceptance rule: when  $|\text{Ared}_k|$  exceeds the noise floor, the ratio  $\text{Ared}_k/\text{Pred}_k$  is informative and standard

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**Algorithm 1** TOBYQA

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**Require:** Oracle  $F(\cdot, \cdot)$ ; starting point  $x_0$ ; initial radius  $\Delta_0$ ; noise level  $\sigma$ .

- 1: Probe  $\{x_0, x_0 \pm \Delta_0 e_j\}_{j=1}^n$  in succession, advancing  $t$  by one unit per query, to build  $\mathcal{Y}_0 = \{(x_i, t_i, F_i)\}_{i=0}^{2n}$ .
  - 2: Estimate the physical-space Hessian diagonal  $\hat{H}_{jj} \leftarrow (F_+^j + F_-^j - 2F_0)/\Delta_0^2$  and seed  $S \leftarrow \text{diag}(1/\sqrt{|\hat{H}_{jj}|})$  (clipped, normalized so that  $\max_j s_j = 1$ ).
  - 3: Infer  $\rho$  from (21); set  $\sigma_{\text{tol}} \leftarrow 4\sigma$ .
  - 4:  $F_{\text{old}} \leftarrow F_0$ ;  $t_{\text{old}} \leftarrow t_0$ ;  $\bar{r} \leftarrow 1$ .
  - 5: **for**  $k = 0, 1, 2, \dots$  **do**
  - 6:    $t \leftarrow t + 1$ .
  - 7:   Update  $S$  by EMA from the diagonal of the previous iteration's recovered Hessian  $G$ .
  - 8:   Assemble  $M$  in physical coordinates from (10) and solve for  $(\lambda, c, g, \gamma)$ .
  - 9:   Solve the hat-space trust-region subproblem by  $\hat{s} \leftarrow \text{TRSAPP}(Sg, \lambda, \{Sd_i\}_{i=1}^m, \Delta_k)$ ; set  $s \leftarrow S\hat{s}$ .
  - 10:   Compute  $\text{Pred}_k$  from (17).
  - 11:   Query  $F_{\text{new}} \leftarrow F(x_k + s, t)$ ;  $\text{Ared}_k \leftarrow F_{\text{old}} - F_{\text{new}} + \gamma(t - t_{\text{old}})$ .
  - 12:   Update  $\Delta_{k+1}$  by the two-regime ratio test of Section 3.2.
  - 13:   Drop sample  $j \leftarrow \arg \min_{i: x_i \neq x_k} t_i$  and replace by the trial point.
  - 14:   **if**  $\text{Ared}_k \geq -\frac{1}{2}\sigma_{\text{tol}}$  **then**
  - 15:     Accept the step:  $x_k \leftarrow x_k + s$ ,  $F_{\text{old}} \leftarrow F_{\text{new}}$ ,  $t_{\text{old}} \leftarrow t$ .
  - 16:   **else if**  $\Delta_k/\Delta_0 < 0.1$  **then**
  - 17:      $t \leftarrow t + 1$ ;  $F_{\text{anchor}} \leftarrow F(x_k, t)$ ; insert at the oldest non-center slot;  $F_{\text{old}} \leftarrow F_{\text{anchor}}$ ,  $t_{\text{old}} \leftarrow t$ .
  - 18:   **end if**
  - 19:   **break** if any termination criterion (T1)–(T4) of Section 3.3 holds.
  - 20: **end for**
  - 21: **return**  $x_k$ .
- 

trust-region radius updates apply; when  $|\text{Ared}_k|$  is comparable to or below the noise floor, the ratio is dominated by the noise term and the radius is adjusted conservatively. The explicit form of both regimes, the smoothing of the ratio across iterations, and the schedule of radius updates are given in Section 3.

Setting  $\gamma = 0$ ,  $\sigma = 0$  and  $\mu \equiv 0$  in (20) recovers the classical actual reduction  $f(x_k) - f(x_k + s)$ , so that  $\text{Ared}_k/\text{Pred}_k$  reduces to the standard trust-region ratio of the noise-free static case.

### 3 The TOBYQA Algorithm

The model-building components of Section 2 are combined into an iterative procedure given as Algorithm 1.

#### 3.1 Initialization

The initial interpolation set  $\mathcal{Y}_0$  is built by an axis probe of length  $\Delta_0$  around the user-supplied starting point  $x_0$ : the algorithm first evaluates the oracle at  $x_0$ , then at  $x_0 \pm \Delta_0 e_j$  for  $j = 1, \dots, n$  in succession, advancing the time stamp by one unit per query. This produces  $m = 2n + 1$  samples in exactly  $2n + 1$  oracle evaluations, and by Corollary 2.8 the augmented constraint matrix  $X_{\text{aug}}$  is of full column rank  $n + 2$  from the first iteration onward. The probe values also yield a finite-difference estimate of the physical-space Hessian diagonal,  $\hat{H}_{jj} \approx (F_+^j + F_-^j - 2F_0)/\Delta_0^2$ , which seeds the affine-scaling matrix  $S$  via  $s_j \propto 1/\sqrt{|\hat{H}_{jj}|}$  before the first trust-region subproblem is solved. The KKT system itself is assembled in physical coordinates throughout, so that  $S$

enters only the trust-region step computation, in accordance with the separation established in Section 2.4.

The remaining tunable parameters are inferred from the same probe data without additional oracle cost. The noise tolerance is set to  $\sigma_{\text{tol}} = 4\sigma$ , where  $\sigma$  is the user-supplied noise level, and the soft-constraint penalty is set to

$$\rho = \text{clip}(10 \text{ spread}_0 / \Delta_0^2, 10, 10^6), \quad \text{spread}_0 := \max_i F_i - \min_i F_i, \quad (21)$$

giving the soft constraint a tenfold safety margin over the empirical curvature scale, with clipping bounds against degenerate flat or extremely steep landscapes.

### 3.2 The main iteration

Each iteration begins by advancing the time stamp by one unit and refreshing the affine-scaling matrix  $S$  via exponential moving average from the diagonal of the previous iteration's recovered Hessian  $G = \sum_i \lambda_i d_i d_i^\top$ . The augmented KKT system (10) is then assembled in physical coordinates and solved as a dense linear system in  $O((m+n)^3) = O(n^3)$  time, returning the joint solution  $(\lambda, c, g, \gamma)$ . The hat-space trust-region subproblem

$$\min_{\hat{s}} (Sg)^\top \hat{s} + \frac{1}{2} \hat{s}^\top \hat{G} \hat{s} \quad \text{subject to} \quad \|\hat{s}\|_2 \leq \Delta_k, \quad (22)$$

with  $\hat{G} = \sum_i \lambda_i (Sd_i)(Sd_i)^\top$ , is solved by the truncated conjugate-gradient method of Powell [1] (the TRSAPP subroutine), which consumes the factored representation of  $\hat{G}$  as a list of vectors  $\{Sd_i\}$  with weights  $\{\lambda_i\}$  and does not assemble  $\hat{G}$  explicitly. The physical step is recovered as  $s = S\hat{s}$ .

The acceptance of the trial point  $x_k + s$  and the update of the trust-region radius are governed by a two-regime ratio test on the drift-compensated reduction  $\text{Ared}_k = F_{\text{old}} - F_{\text{new}} + \gamma \Delta t_q$ , where  $\Delta t_q = t - t_{\text{old}}$  is the elapsed time since the incumbent value  $F_{\text{old}}$  was last refreshed. When  $|\text{Ared}_k|$  is below the noise tolerance  $\sigma_{\text{tol}}$ , the observed reduction is statistically indistinguishable from noise and no instantaneous ratio is computed; the algorithm mildly expands the radius ( $\Delta_{k+1} = \min(1.3 \Delta_k, \Delta_{\text{max}})$ ) when  $\text{Ared}_k \geq -\frac{1}{2}\sigma_{\text{tol}}$  to traverse flat or noisy plateaus, and mildly contracts it ( $\Delta_{k+1} = \max(0.8 \Delta_k, \Delta_{\text{min}})$ ) otherwise. When  $|\text{Ared}_k|$  exceeds  $\sigma_{\text{tol}}$ , the classical ratio  $r_k^{\text{inst}} = \text{Ared}_k / \max(\text{Pred}_k, 10^{-12})$  (clipped to  $[-2, 2]$ ) is smoothed by an exponential moving average  $\bar{r}_k = \beta_k \bar{r}_{k-1} + (1 - \beta_k) r_k^{\text{inst}}$  with iteration-dependent mixing weight  $\beta_k = 0.3 + 0.4 \chi_k$ , where  $\chi_k = \text{clip}(\Delta_k / \Delta_0, 0.05, 1)$  is the relative trust-region scale: a larger  $\Delta_k$  corresponds to an exploratory phase that benefits from a long-memory average, while a smaller  $\Delta_k$  corresponds to a refinement phase that should react quickly to changing local curvature. The radius is then updated by

$$\Delta_{k+1} = \text{clip}(\Delta_k \cdot \phi(\bar{r}_k), \Delta_{\text{min}}, \Delta_{\text{max}}), \quad (23)$$

with scaling factor  $\phi(\bar{r}) = \text{clip}(\exp(\eta(\bar{r} - \bar{r}^*)), 1.1, 2)$  when  $\bar{r} \geq \bar{r}^*$ ,  $\phi(\bar{r}) = 1$  when  $0.1 < \bar{r} < \bar{r}^*$ , and  $\phi(\bar{r}) = \text{clip}(\exp(\eta(\bar{r} - \bar{r}^*)), 0.5, 0.9)$  when  $\bar{r} \leq 0.1$ . Throughout both regimes the accept/reject decision uses the uniform criterion  $\text{Ared}_k \geq -\frac{1}{2}\sigma_{\text{tol}}$ , decoupled from  $\bar{r}_k$ : in a noisy moving-target setting a step that yields a poor instantaneous ratio may nonetheless be a genuine improvement masked by noise, and  $\bar{r}_k$  is appropriate for deciding how aggressively to step next rather than whether to move at all.

At each iteration the trial point enters the interpolation set whether or not the step is accepted, displacing the oldest sample whose spatial location does not coincide with the current center  $x_k$ :

$$j = \arg \min_{i: x_i \neq x_k} t_i. \quad (24)$$

The center exclusion is enforced by assigning an effective time stamp of  $+\infty$  to any sample that coincides with  $x_k$  within machine precision, ensuring that the model always retains an observation

at the incumbent. The simplicity of (24) is enabled by Proposition 2.6: the regularization  $\frac{1}{\rho}I_m$  supplied by the soft constraint makes nonsingularity of  $M$  depend only on the column rank of  $X_{\text{aug}}$ , removing the need for the geometry-improvement machinery of classical NEWUOA-style solvers [1, 3].

A complementary mechanism activates when the trust-region radius has contracted substantially ( $\Delta_k/\Delta_0 < 0.1$ ) and the current step has been rejected. In this regime the recorded incumbent value  $F_{\text{old}}$  may be significantly stale relative to the current time, biasing the next ratio test by an amount the drift correction  $\gamma \Delta t_q$  cannot fully absorb. The algorithm advances time by one unit, re-evaluates  $F$  at the unchanged center  $x_k$ , and replaces  $(F_{\text{old}}, t_{\text{old}})$  with the fresh observation  $(F_{\text{anchor}}, t)$ . The fresh sample also enters the interpolation set at the oldest non-center slot. This time-anchor costs one extra oracle query per triggering iteration and resets the ratio-test reference against current-time noise rather than accumulated drift.

### 3.3 Termination and default parameters

TOBYQA terminates when any of four indicators is satisfied. The trust-region radius indicator (T1) triggers when  $\Delta_k < 10^{-5}$ , at which point the local landscape has been resolved down to numerical precision. The first-order indicator (T2) triggers when the recovered physical gradient and the trust-region radius are both small:  $\|g\|_2 < 10^{-4}$  with  $\Delta_k < 0.05 \Delta_0$  at iterations divisible by 50, or the stricter combination  $\|g\|_2 < 10^{-6}$  with  $\Delta_k < 0.01 \Delta_0$  at every iteration. The stagnation indicator (T3) tracks a counter that increments on rejected low-signal iterations ( $|\text{Ared}_k| \leq \sigma_{\text{tol}}$ ), resets on any accepted iteration with  $\text{Ared}_k > \sigma_{\text{tol}}$ , and decreases by two (clamped at zero) on accepted low-signal iterations; the algorithm terminates when this counter reaches a patience bound  $P_{\text{stag}}$ . The resolution-shrinking indicator (T4) maintains a second counter that tracks consecutive low-signal iterations at the current resolution level  $\rho_{\text{cur}}$ : when this counter reaches  $P_\rho$  and  $\Delta_k < 0.1 \Delta_0$ , both  $\rho_{\text{cur}}$  and  $\Delta_k$  are halved, and the algorithm terminates once  $\rho_{\text{cur}}$  falls below  $10^{-5}$ . The four-criterion test is conservative: in practice (T1) and (T3) account for most terminations on static and dynamic problems respectively, with (T2) and (T4) serving as redundant safeguards.

Table 1: Default parameters of TOBYQA. None of these values is tuned per problem.

| Symbol                | Description                        | Value         |
|-----------------------|------------------------------------|---------------|
| $\Delta_0$            | Initial trust-region radius        | 1.0           |
| $\bar{r}^*$           | Target ratio for radius expansion  | 0.7           |
| $\eta$                | Sensitivity of radius scaling rule | 0.5           |
| $\sigma_{\text{tol}}$ | Noise tolerance                    | $4\sigma$     |
| $P_{\text{stag}}$     | Stagnation patience for (T3)       | $\max(30, n)$ |
| $P_\rho$              | Patience for $\rho$ shrinking (T4) | $\max(20, n)$ |

## 4 Recovery Properties of the Augmented KKT System

The augmented KKT system (10) recovers the model parameters  $(c, g, \gamma)$  as a single linear functional of the data vector  $F$ . This section characterizes that functional and uses it to establish the principal algebraic property of TOBYQA: under additively linear drift, the recovered drift coefficient  $\gamma$  depends on the true drift rate  $\beta$  through a clean additive decomposition that does not contaminate the recovered spatial gradient  $g$ .

## 4.1 The Recovery Operator

Under the column-rank condition of Proposition 2.6, block elimination of  $\lambda$  from (10) and substitution into the constraint rows  $X_{\text{aug}}^\top \lambda = 0$  yields

$$\begin{bmatrix} c \\ g \\ \gamma \end{bmatrix} = \Pi F, \quad \Pi := (X_{\text{aug}}^\top K^{-1} X_{\text{aug}})^{-1} X_{\text{aug}}^\top K^{-1} \in \mathbb{R}^{(n+2) \times m}, \quad (25)$$

where  $K = A + \frac{1}{\rho} I_m$ . We refer to  $\Pi$  as the *recovery operator*; its rows  $\Pi^\top = [\Pi_c^\top, \Pi_g^\top, \Pi_\gamma^\top]$  recover the constant term, the spatial gradient, and the drift coefficient, respectively.

**Lemma 4.1** (Left inverse). *Under the column-rank condition,  $\Pi X_{\text{aug}} = I_{n+2}$ . Reading column by column,*

$$\Pi \mathbf{1}_m = \begin{bmatrix} 1 \\ 0_n \\ 0 \end{bmatrix}, \quad \Pi D_{:,j} = \begin{bmatrix} 0 \\ e_j \\ 0 \end{bmatrix} \quad (j = 1, \dots, n), \quad \Pi \theta = \begin{bmatrix} 0 \\ 0_n \\ 1 \end{bmatrix}. \quad (26)$$

*Proof.*  $\Pi X_{\text{aug}} = (X_{\text{aug}}^\top K^{-1} X_{\text{aug}})^{-1} (X_{\text{aug}}^\top K^{-1} X_{\text{aug}}) = I_{n+2}$ .  $\square$

The third identity in (26) is the algebraic content of the analysis below:  $\Pi_\gamma$  returns one when applied to  $\theta$  and zero when applied to any spatial column of  $X_{\text{aug}}$ .

## 4.2 Additive Decomposition under Linear Drift

We now establish that under linear drift, the recovered  $\gamma$  depends on the true drift rate  $\beta$  through a clean additive decomposition.

**Theorem 4.2** (Additive decomposition under linear drift). *Suppose the observation channel takes the form*

$$F(x, t) = f(x) + \beta t, \quad (27)$$

*with  $\beta \in \mathbb{R}$  a constant drift rate and  $f$  a static objective. Suppose  $X_{\text{aug}}$  satisfies the column-rank condition and  $\rho > 0$ . Define the static-data vector*

$$\tilde{F}_i^{\text{stat}} := f(x_i) + \beta t_k, \quad i = 1, \dots, m, \quad (28)$$

*and let  $(c^{\text{stat}}, g^{\text{stat}}, \gamma^{\text{stat}}) := \Pi \tilde{F}^{\text{stat}}$  denote the KKT solution that would be recovered if  $\tilde{F}^{\text{stat}}$  were the data vector. Then the solution  $(c, g, \gamma)$  of (10) with right-hand side  $F$  satisfies*

$$c = c^{\text{stat}}, \quad g = g^{\text{stat}}, \quad \gamma = \gamma^{\text{stat}} + \beta. \quad (29)$$

*Proof.* For each sample,

$$F_i = f(x_i) + \beta t_i = (f(x_i) + \beta t_k) + \beta (t_i - t_k) = \tilde{F}_i^{\text{stat}} + \beta \theta_i,$$

so  $F = \tilde{F}^{\text{stat}} + \beta \theta$  in vector form. By linearity of  $\Pi$ ,

$$(c, g^\top, \gamma)^\top = \Pi F = \Pi \tilde{F}^{\text{stat}} + \beta \Pi \theta.$$

Lemma 4.1 gives  $\Pi \theta = (0, 0_n^\top, 1)^\top$ , so  $\beta \Pi \theta$  has zero in the  $c$  and  $g$  components and  $\beta$  in the  $\gamma$  component. Combining with  $(c^{\text{stat}}, g^{\text{stat}}, \gamma^{\text{stat}}) = \Pi \tilde{F}^{\text{stat}}$  gives (29).  $\square$

The content of Theorem 4.2 is structural rather than numerical: it states that linear drift contaminates the recovered  $\gamma$  in exactly one way (additively by  $\beta$ ) and does not contaminate the recovered  $c$  or  $g$  at all. The static-data recovery  $(c^{\text{stat}}, g^{\text{stat}}, \gamma^{\text{stat}})$  is itself a property of  $f$  and the sample geometry, independent of  $\beta$ .

In the noiseless static case,  $\gamma^{\text{stat}}$  is not zero in general for finite  $\rho$ : the soft-constraint formulation trades a strictly zero drift estimate against a smaller Frobenius norm of the recovered Hessian, and the magnitude of  $\gamma^{\text{stat}}$  is set by the projection of the quadratic component of  $f$  onto the recovery functional  $\Pi_\gamma$ . The static bias vanishes in the hard-constraint limit  $\rho \rightarrow \infty$  (Proposition 2.9).

**Corollary 4.3** (Exact recovery in the hard-constraint limit). *Under the hypotheses of Theorem 4.2, together with the assumption that  $f$  is quadratic on the sample set, the solution of the hard-constrained augmented system (12) (equivalently, the  $\rho \rightarrow \infty$  limit of (10)) satisfies*

$$\gamma^* = \beta, \quad g^* = \nabla f(x_k), \quad c^* = f(x_k) + \beta t_k.$$

*Proof.* By Proposition 2.9 applied to the static data  $\tilde{F}^{\text{stat}}$  with  $f$  quadratic, the hard-constraint solution recovers  $c^{\text{stat},*} = f(x_k) + \beta t_k$ ,  $g^{\text{stat},*} = \nabla f(x_k)$ ,  $\gamma^{\text{stat},*} = 0$ . Theorem 4.2 extends to the hard-constraint system by the same proof, with  $\Pi$  replaced by its hard-constraint analogue, giving the claim.  $\square$

The drift-compensated ratio test of Section 2.5 is justified by the additive decomposition rather than by exact recovery. To see this, substitute the linear-drift channel (27) and the recovered drift  $\gamma = \gamma^{\text{stat}} + \beta$  into  $\text{Ared}_k = (F_{\text{old}} - F_{\text{new}}) + \gamma \Delta t_q$ :

$$\begin{aligned} \text{Ared}_k &= (f(x_k) + \beta t_{\text{old}}) - (f(x_k + s) + \beta t_{\text{new}}) + (\gamma^{\text{stat}} + \beta) \Delta t_q \\ &= (f(x_k) - f(x_k + s)) + \gamma^{\text{stat}} \Delta t_q. \end{aligned}$$

The true drift contribution  $\beta \Delta t_q$  is removed exactly; the residual term  $\gamma^{\text{stat}} \Delta t_q$  is the same bias that would corrupt the ratio test on a static problem and is independent of  $\beta$ . In particular, in the hard-constraint limit,  $\gamma^{\text{stat}} = 0$  and  $\text{Ared}_k = f(x_k) - f(x_k + s)$  recovers the classical static ratio.

### 4.3 Perturbation under Nonlinear Drift

Theorem 4.2 captures the recovery structure exactly for linear drift. We now quantify how the recovery deviates from this structure when the drift channel  $\mu(t)$  is nonlinear.

**Theorem 4.4** (Recovery bias under nonlinear drift). *Suppose the observation channel takes the additive form*

$$F(x, t) = f(x) + \mu(t) + \varepsilon(x, t), \quad (30)$$

*with  $\mu \in C^2(\mathbb{R}_+)$  a drift function, and  $\varepsilon$  a zero-mean noise term satisfying Assumption 2.3. Suppose  $X_{\text{aug}}$  satisfies the column-rank condition. Let  $(c^{\text{stat}}, g^{\text{stat}}, \gamma^{\text{stat}}) := \Pi \tilde{F}^{\text{stat}}$  be the static-data KKT solution defined as in (28) with  $\beta t_k$  replaced by  $\mu(t_k)$ . Then the solution of (10) with right-hand side  $F$  admits the decomposition*

$$\begin{bmatrix} c \\ g \\ \gamma \end{bmatrix} = \begin{bmatrix} c^{\text{stat}} \\ g^{\text{stat}} \\ \gamma^{\text{stat}} + \mu'(t_k) \end{bmatrix} + \Pi R + \Pi \varepsilon, \quad (31)$$

*where  $R \in \mathbb{R}^m$  has entries  $R_i = \mu(t_i) - \mu(t_k) - \mu'(t_k)(t_i - t_k)$  and satisfies*

$$\|R\|_\infty \leq \frac{1}{2} L_\mu \tau_{\text{max}}^2, \quad L_\mu := \sup_t |\mu''(t)|, \quad \tau_{\text{max}} := \max_i |t_i - t_k|. \quad (32)$$

In particular, the recovered drift coefficient satisfies

$$|\gamma - (\mu'(t_k) + \gamma^{\text{stat}})| \leq \frac{\sqrt{m}}{2} \|\Pi_\gamma\|_2 L_\mu \tau_{\max}^2 + |\Pi_\gamma \varepsilon|, \quad (33)$$

and the recovered gradient satisfies

$$\|g - g^{\text{stat}}\|_2 \leq \frac{\sqrt{m}}{2} \|\Pi_g\|_2 L_\mu \tau_{\max}^2 + \|\Pi_g \varepsilon\|_2. \quad (34)$$

*Proof.* Taylor expansion of  $\mu$  around  $t_k$  gives, for each sample,  $\mu(t_i) = \mu(t_k) + \mu'(t_k) \theta_i + R_i$  with  $|R_i| \leq \frac{1}{2} \sup_\xi |\mu''(\xi)| \theta_i^2 \leq \frac{1}{2} L_\mu \tau_{\max}^2$ . In vector form,  $F = \tilde{F}^{\text{stat}} + \mu'(t_k) \theta + R + \varepsilon$ . Applying  $\Pi$  and using  $\Pi \theta = (0, 0_n^\top, 1)^\top$  from Lemma 4.1 gives (31). The component bounds (33)–(34) follow by the operator-norm inequality  $\|\Pi_\gamma R\| \leq \|\Pi_\gamma\|_2 \|R\|_2$  and analogously for  $\Pi_g$ , with  $\|R\|_2 \leq \sqrt{m} \|R\|_\infty$ .  $\square$

The decomposition (31) exposes the structure of the recovery under nonlinear drift. The deterministic part of the bias consists of two pieces: the static bias  $(c^{\text{stat}}, g^{\text{stat}}, \gamma^{\text{stat}})$  that would already be present on a drift-free problem at finite  $\rho$ , and a curvature-of-drift term proportional to  $L_\mu \tau_{\max}^2$ . The recovered drift coefficient  $\gamma$  captures the *instantaneous* drift rate  $\mu'(t_k)$  at the current reference time, up to these two biases and the noise contribution  $\Pi \varepsilon$ .

Two algorithmic implications follow. First, the curvature-of-drift bias scales with the square of the maximum sample age  $\tau_{\max}$ . The time-priority replacement rule (24) keeps  $\tau_{\max}$  bounded by the interpolation set size  $m = 2n + 1$ , and the time-anchor mechanism of Section 3.2 refreshes the reference sample whenever the trust-region radius has contracted significantly, controlling  $\tau_{\max}$  further in late iterations. Both mechanisms are motivated directly by (33)–(34). Second, the bound (33) grows with  $L_\mu$ . On environments where  $L_\mu$  is large—periodic or chaotic drift with rapid temporal curvature—the recovery of  $\gamma$  deteriorates accordingly, predicting a narrowing of TOBYQA’s advantage over static-objective trust-region baselines on such environments. This is consistent with the empirical pattern observed in Section 5.

**Remark 4.5.** The additive form (30) corresponds to the special case  $\phi(v, t) = v + \mu(t)$  of the monotone channel (2). For a general monotone  $\phi(v, t)$ , expanding  $\phi(f(x), t)$  around  $(f(x_k), t_k)$  produces, in addition to the additive drift handled here, a multiplicative factor  $\partial_v \phi(f(x_k), t_k) > 0$  on the linear spatial term and a space-time cross term proportional to  $\partial_{vt} \phi(f(x_k), t_k)$ . The multiplicative factor rescales the recovered gradient but does not change its direction; the cross term, in contrast, lies in neither the column span of  $D$  nor that of  $\theta$ , and contributes a bias to the recovered  $g$  whose direction depends on  $\partial_{vt} \phi$  and the sample geometry. A quantitative bound on the cross-term contribution requires structural assumptions on  $\phi$  beyond strict monotonicity in  $v$  and is left to future work. The strict monotonicity assumption of Section 2.1 ensures only that the global minimizer of  $F(\cdot, t)$  coincides with that of  $f$  at every  $t$ , not that local first-order recovery is unbiased.

#### 4.4 Conditional Convergence under a Geometric Assumption

The recovery results of §4.2 and §4.3 hold for any column-rank  $X_{\text{aug}}$ . To pass from per-iteration recovery to a convergence statement on  $\{x_k\}$ , one needs a quantitative geometric condition of the kind that classical derivative-free trust-region analyses [16] guarantee through  $\Lambda$ -poisedness maintenance. As discussed in Remark 2.7, the soft constraint  $\frac{1}{\rho} I_m$  in TOBYQA reduces nonsingularity of  $M$  to the linear independence of the columns of  $X_{\text{aug}}$  without enforcing  $\Lambda$ -poisedness, and the time-priority replacement rule (24) does not actively maintain geometry. We therefore state convergence conditionally on the geometric quality of the sample sequence, and verify the condition empirically in §4.4. The argument follows the trust-region convergence framework of [16] with three substitutions: (i) the fully-linear class of [16, Definition 3.1] is replaced by the fully-quadratic recovery bound provided by our Lemma 4.8 below, (ii) the

$\Lambda$ -poisedness-based model maintenance of [16, §4] is replaced by Assumption 4.6, and (iii) the criticality step of [16, Algorithm 4.1, Step 1] is replaced by a noise-floor analysis that yields a neighborhood convergence statement  $\liminf \|\nabla f(x_k)\| \leq C\sigma$  in place of their zero-limit.

**Assumption 4.6** (Sustained geometric condition). *There exist constants  $\eta > 0$  and  $K_0 \in \mathbb{N}$  such that*

$$\sigma_{\min}(X_{\text{aug}}^{(k)}) \geq \eta \Delta_k \quad \forall k \geq K_0, \quad (35)$$

where  $X_{\text{aug}}^{(k)} \in \mathbb{R}^{m \times (n+2)}$  is the augmented constraint matrix (11) at iteration  $k$ .

The constant  $\eta$  plays the role that  $\Lambda$  plays in the  $\Lambda$ -poisedness analyses of [16]: it controls the conditioning of the recovery operator (25) through the smallest singular value of  $X_{\text{aug}}$ , rather than through the  $L^\infty$  norm of the Lagrange interpolation basis. By Corollary 2.8, Assumption 4.6 holds at  $k = 0$  with  $\eta = 1$  for the axis-probe initialization.

### A fully-quadratic recovery bound

We begin with an operator-norm bound on the recovery map (25).

**Lemma 4.7** (Recovery-operator bound). *Under Assumption 4.6, for every  $k \geq K_0$ ,*

$$\|\Pi_k\|_{\text{op}} \leq \frac{(\frac{m}{2}\Delta_k^4 + 1/\rho)\rho\|X_{\text{aug}}^{(k)}\|_{\text{op}}}{\eta^2\Delta_k^2}. \quad (36)$$

For  $\Delta_k \leq \Delta_0$  this is bounded above by  $C_\Pi(\rho, \eta, m, \Delta_0)/\Delta_k^2$  for a constant  $C_\Pi$  depending only on its arguments.

*Proof.* With  $K = A + \frac{1}{\rho}I_m$  from (25), submultiplicativity gives

$$\|\Pi_k\|_{\text{op}} \leq \|(X_{\text{aug}}^{(k)\top} K^{-1} X_{\text{aug}}^{(k)})^{-1}\|_{\text{op}} \cdot \|X_{\text{aug}}^{(k)}\|_{\text{op}} \cdot \|K^{-1}\|_{\text{op}}.$$

Since  $K \succeq (1/\rho)I_m$ ,  $\|K^{-1}\|_{\text{op}} \leq \rho$ . The entry-wise bound  $A_{ij} = \frac{1}{2}(d_i^\top d_j)^2 \leq \frac{1}{2}\Delta_k^4$  gives  $\|K\|_{\text{op}} \leq \frac{m}{2}\Delta_k^4 + 1/\rho$ . For the inner inverse,  $X_{\text{aug}}^{(k)\top} K^{-1} X_{\text{aug}}^{(k)} \succeq \|K\|_{\text{op}}^{-1} X_{\text{aug}}^{(k)\top} X_{\text{aug}}^{(k)} \succeq \sigma_{\min}(X_{\text{aug}}^{(k)})^2 / \|K\|_{\text{op}} \cdot I$ , which under Assumption 4.6 yields  $\|(X_{\text{aug}}^{(k)\top} K^{-1} X_{\text{aug}}^{(k)})^{-1}\|_{\text{op}} \leq \|K\|_{\text{op}} / (\eta^2 \Delta_k^2)$ . Combining gives (36).  $\square$

The next lemma is the central recovery estimate.

**Lemma 4.8** (Fully-quadratic recovery). *Suppose Assumptions 2.1, 2.2, 2.3, and 4.6 hold, and that the observation channel has the additive form  $F(x, t) = f(x) + \mu(t) + \varepsilon(x, t)$  with  $\mu \in C^2$  and  $|\mu''| \leq L_\mu$ . Then for every  $k \geq K_0$  the recovered gradient  $g_k$  satisfies*

$$\|g_k - \nabla f(x_k)\| \leq C_g (L_g \Delta_k^2 + L_H \Delta_k^3 + L_\mu \tau_{\max, k}^2) + \|\Pi_{g, k}\|_{\text{op}} \|\varepsilon_k\|, \quad (37)$$

where  $\Pi_{g, k} \in \mathbb{R}^{n \times m}$  is the  $g$ -block of  $\Pi_k$ ,  $\tau_{\max, k} := \max_i |t_i - t_k|$  over the current interpolation set,  $\varepsilon_k \in \mathbb{R}^m$  collects the noise realizations at the sample times, and the constant  $C_g = C_g(\rho, \eta, m, \Delta_0)$  depends only on its arguments.

*Proof.* Decompose the observation vector along the Taylor expansion of  $f$  at  $x_k$  and the chord of  $\mu$  at  $t_k$ . With  $d_i = x_i - x_k$  and  $\theta_i = t_i - t_k$ ,

$$F_i = (f(x_k) + \mu(t_k)) + \nabla f(x_k)^\top d_i + \underbrace{\frac{1}{2} d_i^\top \nabla^2 f(x_k) d_i}_{=: q_i} + \underbrace{\mu'(t_k) \theta_i}_{\text{cubic}} + \underbrace{R_i}_{\mu\text{-residual}} + \varepsilon_i,$$

where  $|r_i| \leq \frac{1}{6}L_H\|d_i\|^3 \leq \frac{1}{6}L_H\Delta_k^3$  by Assumption 2.1 and  $|R_i| = |\mu(t_i) - \mu(t_k) - \mu'(t_k)\theta_i| \leq \frac{1}{2}L_\mu\tau_{\max,k}^2$  by the assumption on  $\mu$ . In matrix form,

$$F = (f(x_k) + \mu(t_k))\mathbf{1}_m + D\nabla f(x_k) + q + r + \mu'(t_k)\theta + R + \varepsilon.$$

Apply  $\Pi_{g,k}$  (the  $g$ -block of  $\Pi_k$ ). By Lemma 4.1,  $\Pi_k\mathbf{1}_m = [1, 0_n, 0]^\top$ ,  $\Pi_k D_{:,j} = [0, e_j, 0]^\top$ , and  $\Pi_k\theta = [0, 0_n, 1]^\top$ , so the  $g$ -block satisfies  $\Pi_{g,k}\mathbf{1}_m = 0$ ,  $\Pi_{g,k}D = I_n$ , and  $\Pi_{g,k}\theta = 0$ . Consequently

$$g_k = \Pi_{g,k}F = \nabla f(x_k) + \Pi_{g,k}q + \Pi_{g,k}r + \Pi_{g,k}R + \Pi_{g,k}\varepsilon_k.$$

Term-wise bounds:  $\|q\| \leq \frac{\sqrt{m}}{2}L_g\Delta_k^2$ ,  $\|r\| \leq \frac{\sqrt{m}}{6}L_H\Delta_k^3$ , and  $\|R\| \leq \frac{\sqrt{m}}{2}L_\mu\tau_{\max,k}^2$ . The first three contributions to  $\|g_k - \nabla f(x_k)\|$  are therefore bounded by  $\|\Pi_{g,k}\|_{\text{op}}$  times the corresponding norms; Lemma 4.7 combined with  $\|\Pi_{g,k}\|_{\text{op}} \leq \|\Pi_k\|_{\text{op}}$  absorbs the deterministic terms into the constant  $C_g$ . The temporal column  $\theta$  contributes nothing to the bias of  $g$  because  $\Pi_{g,k}\theta = 0$ ; its only effect is on the  $(n+2)$ -column structure of  $X_{\text{aug}}$ , which is built into Assumption 4.6.  $\square$

The deterministic part of (37) is  $O(\Delta_k^2)$ , strictly stronger than the fully-linear  $O(\Delta_k)$  required by [16, Definition 3.1]. The deterministic constant  $C_gL_g$  thus plays the role of their  $\kappa_{\text{eg}}$  multiplied by  $\Delta_k$ , while the stochastic term  $\|\Pi_{g,k}\|_{\text{op}}\|\varepsilon_k\|$  is the extension specific to the noise model in Assumption 2.3, which is outside the deterministic scope of [16]. An analogous bound on the recovered Hessian  $G_k = \sum_i \lambda_i d_i d_i^\top$  follows from applying the same Taylor decomposition to the  $\lambda$ -block of  $\Pi_k$ ; we record only the consequence that suffices for what follows:

$$\|G_k\|_{\text{op}} \leq \kappa_{\text{bhm}} \quad (38)$$

for a constant  $\kappa_{\text{bhm}} = \kappa_{\text{bhm}}(\rho, \eta, m, \Delta_0, L_g, L_H, L_\mu, \tau_{\max}, \sigma)$ , which is the analogue of [16, Assumption 5.2] and holds in expectation under Assumptions 2.1–4.6.

### Model Cauchy decrease

The hat-space subproblem (14) is solved by the truncated conjugate-gradient routine of [1], which is known to achieve a fixed fraction of the Cauchy decrease.

**Lemma 4.9** (Fraction of Cauchy decrease). *Let  $\hat{s}_k$  be the output of the truncated-CG solver TRSAPP applied to (14), with  $\hat{g}_k = Sg_k$ ,  $\hat{G}_k = SG_kS$ , and  $\|\hat{s}_k\| \leq \Delta_k$ . The predicted reduction (17) satisfies*

$$\text{Pred}_k \geq \frac{\kappa_{\text{fcd}}}{2} \|\hat{g}_k\| \min\left[\frac{\|\hat{g}_k\|}{\|\hat{G}_k\|_{\text{op}}}, \Delta_k\right], \quad (39)$$

for some  $\kappa_{\text{fcd}} \in (0, 1]$  depending only on the truncated-CG termination criterion.

*Proof.* The truncated-CG solver [1] terminates at a point satisfying the fraction-of-Cauchy-decrease condition [16, Assumption 2.1] applied to the hat-space model  $\hat{g}_k^\top \hat{s} + \frac{1}{2}\hat{s}^\top \hat{G}_k \hat{s}$  on  $\|\hat{s}\| \leq \Delta_k$ . Then [16, Theorem 2.1 and (2.6)] yield (39) for the hat-space predicted reduction  $-\hat{g}_k^\top \hat{s}_k - \frac{1}{2}\hat{s}_k^\top \hat{G}_k \hat{s}_k$ . The identity  $-\hat{g}_k^\top \hat{s}_k - \frac{1}{2}\hat{s}_k^\top \hat{G}_k \hat{s}_k = -\hat{g}_k^\top \hat{s}_k - \frac{1}{2}\hat{s}_k^\top \hat{G}_k \hat{s}_k$  follows from  $s_k = S\hat{s}_k$  and (13); the same quantity equals (17) after substituting  $G_k = \sum_i \lambda_i d_i d_i^\top$ .  $\square$

By the affine-scaling normalization (16),  $\max_j (S_k)_{jj} = 1$ ; let  $s_{\min,k} := \min_j (S_k)_{jj} > 0$  and write  $s_{\min} := \inf_k s_{\min,k}$ , which is positive under the boundedness of  $|G_{jj}|$  implied by (38). Then  $\|\hat{g}_k\| \geq s_{\min}\|g_k\|$  and  $\|\hat{G}_k\|_{\text{op}} \leq \kappa_{\text{bhm}}$ , so when  $\Delta_k \leq s_{\min}\|g_k\|/\kappa_{\text{bhm}}$  the minimum in (39) is  $\Delta_k$ , giving

$$\text{Pred}_k \geq \frac{\kappa_{\text{fcd}} s_{\min}}{2} \|g_k\| \Delta_k. \quad (40)$$

### Step acceptance under the noise floor

We now establish a noise-extended analogue of [16, Lemma 5.2]: under sufficient agreement between  $\text{Pred}_k$  and  $\text{Ared}_k$ , the TOBYQA acceptance criterion  $\text{Ared}_k \geq -\frac{1}{2}\sigma_{\text{tol}}$  holds.

**Lemma 4.10** (Step acceptance). *Under Assumptions 2.1–4.6, suppose the observation channel has the additive form  $F(x, t) = f(x) + \mu(t) + \varepsilon(x, t)$ . There exist constants  $\bar{c}_1, \bar{c}_2, \bar{c}_3$  (depending on the constants of Lemmas 4.8 and 4.9 and on  $\eta_1 \in (0, 1)$ ) such that on the event*

$$\|\Pi_{g,k}\varepsilon_k\| \leq \bar{c}_1\sigma, \quad |\varepsilon_k^{\text{old}} - \varepsilon_k^{\text{new}}| \leq \bar{c}_2\sigma, \quad (41)$$

*which holds with probability at least  $1 - \delta_k$  for  $\delta_k = \delta_k(\bar{c}_1, \bar{c}_2, \Delta_0, m)$  given by Chebyshev's inequality, the following implication is valid: if*

$$\Delta_k \leq \bar{c}_3 \|g_k\| \quad \text{and} \quad \|g_k\| \Delta_k \geq \sigma_{\text{tol}}, \quad (42)$$

*then iteration  $k$  is successful in the sense of [ , Algorithm 6.1] (Algorithm 1 of this paper):  $\text{Ared}_k \geq -\frac{1}{2}\sigma_{\text{tol}}$ .*

*Proof.* By Proposition 2.12, under the additive observation channel

$$\text{Ared}_k - \text{Pred}_k = [f(x_k) - f(x_k + s_k)] - \text{Pred}_k + [\gamma_k \Delta t_q - (\mu(t_{\text{new}}) - \mu(t_{\text{old}}))] + (\varepsilon_k^{\text{old}} - \varepsilon_k^{\text{new}}).$$

The first bracket is bounded by Taylor's theorem and Lemma 4.8: writing

$$f(x_k + s_k) = f(x_k) + \nabla f(x_k)^\top s_k + \frac{1}{2} s_k^\top \nabla^2 f(\xi_k) s_k$$

for some  $\xi_k$  on the segment and  $\text{Pred}_k = -g_k^\top s_k - \frac{1}{2} s_k^\top G_k s_k$  (using  $G_k = \sum_i \lambda_i d_i d_i^\top$  in (17)), we have

$$|[f(x_k) - f(x_k + s_k)] - \text{Pred}_k| \leq \|\nabla f(x_k) - g_k\| \|s_k\| + \frac{1}{2} \|\nabla^2 f(\xi_k) - G_k\|_{\text{op}} \|s_k\|^2.$$

By Lemma 4.8 on the event (41) and the Hessian-recovery analogue producing  $\|\nabla^2 f(\xi_k) - G_k\|_{\text{op}} \leq C_G(L_g + L_H D \Delta_k + L_\mu \tau_{\text{max},k}^2 / \Delta_k^2)$  with constant  $C_G$  from the same Taylor expansion applied to the  $\lambda$ -block of  $\Pi_k$ , both terms are absorbed into  $\bar{c}_4(L_g \Delta_k^2 + L_H \Delta_k^3 + L_\mu \tau_{\text{max},k}^2 + \bar{c}_1 \sigma \Delta_k)$  for a constant  $\bar{c}_4$ . The drift-residual bracket vanishes in the linear-drift case by Theorem 4.2 and is otherwise bounded by  $\frac{1}{2} L_\mu \tau_{\text{max},k}^2$  via the cited Taylor estimate on  $\mu$ . The third bracket is at most  $\bar{c}_2 \sigma$  on (41). Combining and using the Cauchy lower bound (40), the assumption  $\|g_k\| \Delta_k \geq \sigma_{\text{tol}}$ , and  $\sigma_{\text{tol}} = 4\sigma$ , the deterministic part of  $|\text{Ared}_k - \text{Pred}_k|$  is at most  $\frac{1}{4} \text{Pred}_k$  once  $\Delta_k \leq \bar{c}_3 \|g_k\|$  with  $\bar{c}_3 = \bar{c}_3(\bar{c}_4, \kappa_{\text{fcd}}, s_{\text{min}}, L_g, L_H, L_\mu, \tau_{\text{max}})$ , and the noise part is at most  $\frac{1}{4} \text{Pred}_k$  by (42). Hence  $\text{Ared}_k \geq \frac{1}{2} \text{Pred}_k > 0 \geq -\frac{1}{2} \sigma_{\text{tol}}$ , so the acceptance criterion of (1) is satisfied.

The probability statement on (41) follows from Chebyshev applied to  $\mathbb{E}[\|\Pi_{g,k}\varepsilon_k\|^2] \leq m\sigma^2 \|\Pi_{g,k}\|_{\text{op}}^2$  (using Assumption 2.3 for independence and zero-mean) and the analogous bound for the noise-difference term.  $\square$

### Trust-region radius dynamics

**Lemma 4.11** (Radius does not collapse above the noise floor). *Under the hypotheses of Lemma 4.10, suppose there exist  $\kappa_4 > 0$  and an infinite index set  $\mathcal{K} \subset \mathbb{N}$  such that  $\|g_k\| \geq \kappa_4$  for all  $k \in \mathcal{K}$  and  $\kappa_4 \Delta_0 \geq \sigma_{\text{tol}}$ . Then there exists  $\kappa_5 > 0$  such that  $\Delta_k \geq \kappa_5$  for all  $k \in \mathcal{K}$  with probability at least  $1 - \sum_{k \in \mathcal{K}} \delta_k$ , where  $\delta_k$  is from Lemma 4.10.*

*Proof.* This is the noise-extended analogue of [16, Lemma 5.3], with the criticality-step lower bound  $\beta \|g_k\|$  from their Algorithm 4.1, Step 1, removed. By Lemma 4.10, on the intersection of the events (41) over  $k \in \mathcal{K}$ , whenever  $\Delta_k \leq \bar{c}_3 \kappa_4$  iteration  $k$  is successful. By the radius-update rule (23) of (1), successful iterations with sufficiently large smoothed ratio  $\bar{r}_k \geq \bar{r}^*$  produce

$\Delta_{k+1} \geq \Delta_k$ . The product of the contraction factor  $\gamma \in (0.5, 0.9]$  across unsuccessful iterations and the threshold  $\bar{c}_3 \kappa_4$  gives the lower bound  $\kappa_5 = \gamma \bar{c}_3 \kappa_4$ .

We note that this argument is structurally identical to the proof of [16, Lemma 5.3], with two substitutions: the trigger for successful iterations is our Lemma 4.10 in place of their [16, Lemma 5.2], and the  $\beta \|g_k\|$  lower bound from their criticality step is absent, leaving the radius-update rule of (1) as the sole mechanism. Both changes only affect the value of  $\kappa_5$  and not its positivity.  $\square$

### Neighborhood convergence theorem

**Theorem 4.12** (Conditional neighborhood convergence). *Suppose Assumptions 2.1, 2.2, 2.3, and 4.6 hold, and that the observation channel has the additive form  $F(x, t) = f(x) + \mu(t) + \varepsilon(x, t)$  with  $\mu \in C^2$  and  $|\mu''| \leq L_\mu$ . There exists a constant  $C_\star = C_\star(\rho, \eta, m, n, L_g, L_H, L_\mu, \tau_{\max}, \kappa_{\text{bhm}}, \kappa_{\text{fcd}}, s_{\min})$  such that, with probability tending to one over the noise sequence  $\{\varepsilon_k\}_{k \geq K_0}$ ,*

$$\liminf_{k \rightarrow \infty} \|\nabla f(x_k)\| \leq C_\star(\sigma + L_\mu \tau_{\max}^2). \quad (43)$$

*Proof.* Suppose for contradiction that there exists  $\epsilon_0 > 0$  and  $\bar{K} \geq K_0$  such that  $\|\nabla f(x_k)\| \geq C_\star(\sigma + L_\mu \tau_{\max}^2) + \epsilon_0$  for all  $k \geq \bar{K}$ . Provided  $C_\star$  is chosen at least  $C_g$  in Lemma 4.8 divided by the worst-case affine-scaling factor and times a margin of 2, the bound (37) on the event (41) gives  $\|g_k\| \geq \epsilon_0/2$  for all such  $k$ .

By Lemma 4.11 applied with  $\kappa_4 = \epsilon_0/2$ , we have  $\Delta_k \geq \kappa_5 > 0$  for all  $k \geq \bar{K}'$  for some  $\bar{K}' \geq \bar{K}$ , with the same probability. Therefore (42) of Lemma 4.10 is satisfied for all  $k \geq \bar{K}''$  for some  $\bar{K}''$ , since  $\Delta_k \leq \Delta_0$  and the choice of  $C_\star$  ensures  $\|g_k\| \Delta_k \geq \sigma_{\text{tol}}$ . Hence each such iteration is successful and satisfies, by the proof of Lemma 4.10 together with (40),

$$\text{Ared}_k \geq \frac{1}{2} \text{Pred}_k \geq \frac{\kappa_{\text{fcd}} s_{\min}}{4} \|g_k\| \Delta_k \geq \frac{\kappa_{\text{fcd}} s_{\min} \epsilon_0 \kappa_5}{8}. \quad (44)$$

By (20) of Proposition 2.12 on the event (41) of Lemma 4.10,

$$f(x_k) - f(x_{k+1}) = \text{Ared}_k - [\gamma_k \Delta t_q - (\mu(t_{\text{new}}) - \mu(t_{\text{old}}))] - (\varepsilon_k^{\text{old}} - \varepsilon_k^{\text{new}}) \geq \frac{\kappa_{\text{fcd}} s_{\min} \epsilon_0 \kappa_5}{8} - \bar{c}_5(\sigma + L_\mu \tau_{\max}^2),$$

and an appropriate choice of  $C_\star$  makes the right-hand side at least  $\frac{1}{16} \kappa_{\text{fcd}} s_{\min} \epsilon_0 \kappa_5$ . Summing this bound over  $k \geq \bar{K}''$  contradicts the lower bound  $f \geq f_{\text{low}}$  of Assumption 2.1. Therefore the assumption  $\|\nabla f(x_k)\| \geq C_\star(\sigma + L_\mu \tau_{\max}^2) + \epsilon_0$  must fail for some  $k \geq \bar{K}$  (in fact, for infinitely many  $k$ ), which gives (43).

This argument mirrors the proof of [16, Theorem 5.9], with two modifications. First, we use the additive decomposition (20) of Proposition 2.12 to convert  $\text{Ared}_k$  into a function-value decrease modulo a noise-and-drift residual, in place of their direct use of  $f(x_k) - f(x_{k+1}) \geq \eta_1[\text{model decrease}]$ . Second, we do not invoke the bridge from  $\|g_k\|$  to  $\|\nabla f(x_k)\|$  through the criticality-step coupling  $\Delta_k \leq \mu \|g_k\|$  used in [16, Lemma 5.7]; the bridge here is provided directly by Lemma 4.8, and yields the neighborhood statement (43) rather than the zero-limit  $\liminf \|\nabla f\| = 0$  of [16, Theorem 5.8].  $\square$

The theorem says: the iterates accumulate near the set of  $\nabla f$ -critical points up to an accuracy floor proportional to the noise level  $\sigma$  and to the squared time span  $\tau_{\max}^2$  of the interpolation window, the latter weighted by the second-order temporal smoothness  $L_\mu$  of the drift. In the linear-drift case  $\mu(t) = \beta t$ ,  $L_\mu = 0$  and the bound reduces to  $C_\star \sigma$ . In the noise-free linear-drift case  $\sigma = 0 = L_\mu$  the bound is trivial and recovers the classical  $\liminf \|\nabla f\| = 0$  conclusion of [16], modulo the geometric Assumption 4.6.

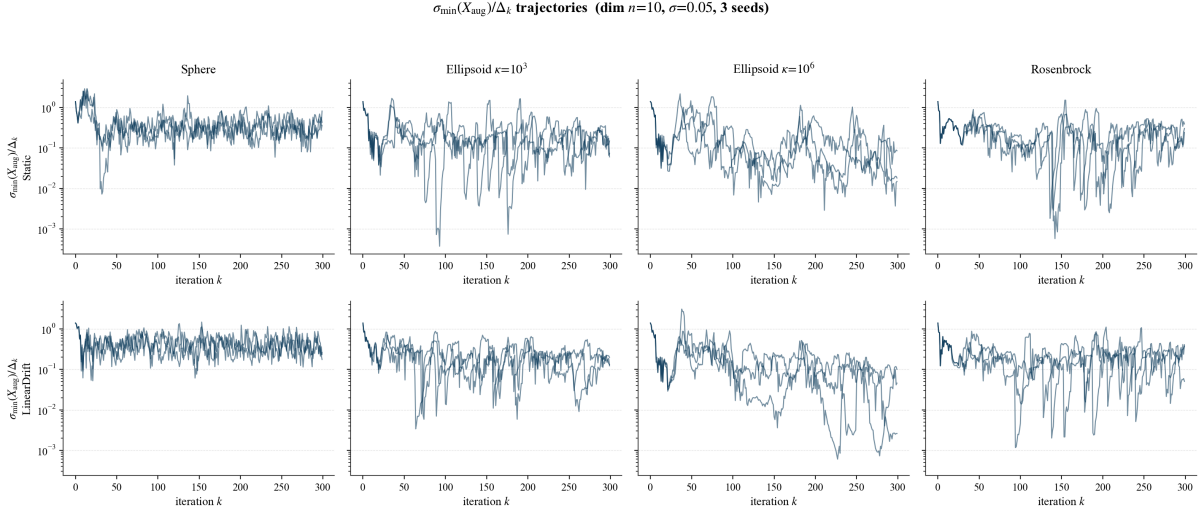


Figure 1: Empirical verification of Assumption 4.6: the ratio  $\sigma_{\min}(X_{\text{aug}}^{(k)})/\Delta_k$  along TOBYQA trajectories. Four representative problems (sphere, ellipsoid  $\kappa = 10^3$ , ellipsoid  $\kappa = 10^6$ , Rosenbrock),  $n = 10$ ,  $\sigma = 0.05$ , three seeds each, in two environments. The ratio remains bounded with median 0.21 and fifth percentile 0.017 across 25,200 iterations.

### Empirical verification of Assumption 4.6

The validity of Theorem 4.12 rests on Assumption 4.6. Since the time-priority replacement rule (24) does not enforce this condition by construction, we verify it empirically. Figure 1 reports  $\sigma_{\min}(X_{\text{aug}}^{(k)})/\Delta_k$  along TOBYQA trajectories on four representative problems (isotropic and anisotropic ellipsoids with condition numbers up to  $10^6$  and the nonconvex Rosenbrock function) under both Static and LinearDrift environments. Across 84 trajectories totaling 25,200 iterations, the ratio has median 0.21 and fifth percentile 0.017, and shows no monotone-decay signature: the trust-region dynamics maintain the ratio within a bounded range without an explicit geometry-improvement step. While this does not constitute a proof of Assumption 4.6, the empirical behavior supports its validity on the problem classes we test.

## 5 Experimental Evaluation

We evaluate TOBYQA on three benchmarks. The Moré–Wild test suite of 88 smooth unconstrained functions [12], wrapped in seven temporal environments and run at three problem dimensions, gives the main comparison against representative derivative-free baselines. A 29-function synthetic suite, stratified into six curvature regimes, supports a controlled ablation that isolates the contribution of the two principal ingredients of TOBYQA: the augmented KKT system with drift unknown and the affine-scaling subproblem. A hyperparameter sensitivity study, run on a representative subset of the synthetic suite, characterizes the robustness of the algorithm to its two user-facing parameters, the soft-constraint penalty  $\rho$  and the initial trust-region radius  $\Delta_0$ .

### 5.1 Experimental Setup

The Moré–Wild suite consists of 88 smooth unconstrained problems drawn from the CUTEst collection. We restrict to problems compatible with each dimension  $n \in \{6, 10, 20\}$ , use the standard starting points  $x_0 = \text{linspace}(-1.5, 1.2, n)$ , and run 5 random seeds per instance. The synthetic suite of 29 functions is partitioned by curvature structure into six groups: isotropic convex (4 problems), anisotropic convex (5), coupled convex (5), weakly nonconvex (5), perturbed

Table 2: Temporal environments used in the Moré–Wild benchmark. Each wraps the static objective  $f(x)$  as  $F(x, t) = E(f(x), x, t) + \mathcal{N}(0, \sigma^2)$ .

| Environment  | Form $E(f, x, t)$                                       | Drift type               |
|--------------|---------------------------------------------------------|--------------------------|
| Static       | $f(x)$                                                  | none                     |
| LinearDrift  | $f(x) + 2t$                                             | additive linear          |
| MultCoupling | $f(x) (1 + 0.005 t)$                                    | multiplicative linear    |
| PeriodicAdd  | $f(x) + 5 \sin(0.05 t)$                                 | additive periodic        |
| PeriodicMult | $f(x) (1 + 0.5 \sin(0.02 t))$                           | multiplicative periodic  |
| ChaoticDrift | $f(x) + 5 \sin(0.07 t) + 2.5 \cos(0.13 t)$              | multi-frequency          |
| Hetero       | $f(x) + 2t + \mathcal{N}(0, \sigma^2(1 + 0.3 \ x\ )^2)$ | linear + heteroscedastic |

convex (5), and translated/rotated (5). The stratification allows per-group analysis of which components of TOBYQA contribute on which curvature regime.

Each problem  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is wrapped in seven temporal environments listed in Table 2. *Static* preserves the original objective and serves as the no-drift baseline; the remaining six environments span the drift taxonomy from purely additive linear drift, the regime in which Theorem 4.2 predicts an exact additive recovery, to chaotic and heteroscedastic perturbations that violate the algorithmic ansatz. Each environment adds i.i.d. Gaussian observation noise of standard deviation  $\sigma = 0.1$  in the main benchmark.

We compare TOBYQA against five baselines spanning the dominant classes of derivative-free optimization: NEWUOA [1], the canonical trust-region quadratic-interpolation method on which TOBYQA builds; BOBYQA [5], Powell’s bound-constrained extension applied here without bounds; DFO-LS [6], a modern noise-aware trust-region DFO method with its noise-handling option enabled; NMSMAX, an adaptive Nelder–Mead variant representing the direct-search class; and Fminunc (BFGS) via SciPy [13] with finite-difference gradients, included to delineate the failure mode of derivative-based methods on noisy time-varying problems.

Each algorithm receives a budget of 1500 function evaluations on every (problem, environment, seed, dimension) instance. To compare algorithms on a common metric, every trajectory point is re-evaluated on the noise-free static  $f$ , so the reported optimality gap  $\min_t f(x_t) - f^*$  measures how well each algorithm tracks the underlying minimizer. We summarize results through the average rank across (problem, seed) pairs, the geometric-mean optimality gap in  $\log_{10}$  scale, and the Moré–Wild data profile [12] that gives the fraction of problems solved to tolerance  $10^{-1}$  as a function of the simplex-gradient budget  $\kappa = N_f/(n + 1)$ .

## 5.2 Main Benchmark Results

Figures 2–4 report the Moré–Wild benchmark at  $n = 6$ ,  $n = 10$  and  $n = 20$ . TOBYQA is the top-ranked method on every (dimension, environment) cell tested, with average rank ranging from 1.31 at the most favorable cell ( $n = 6$ , LinearDrift) to 2.52 at the least favorable ( $n = 20$ , Static). DFO-LS holds the second rank on most environments and slightly outperforms NEWUOA in mean  $\log_{10}$  optimality gap. The TOBYQA advantage is modest on Static—between 0.2 and 0.4 orders of magnitude over the second-place baseline, regardless of dimension—and substantial on the additive-linear-drift environment, where it reaches 1.5 orders of magnitude. The margin narrows on environments whose drift is nonlinear in time. We unpack the pattern by dimension below.

At  $n = 6$  (Figure 2), TOBYQA achieves average rank 2.02 on Static and between 1.31 and 1.97 on the six drift environments. In mean  $\log_{10}$  gap, the Static column reads  $-3.16$  for TOBYQA against  $-2.87$  for DFO-LS and  $-2.59$  for NEWUOA, an advantage of 0.3 orders of magnitude that is indistinguishable from the trust-region cluster at the low end of the budget

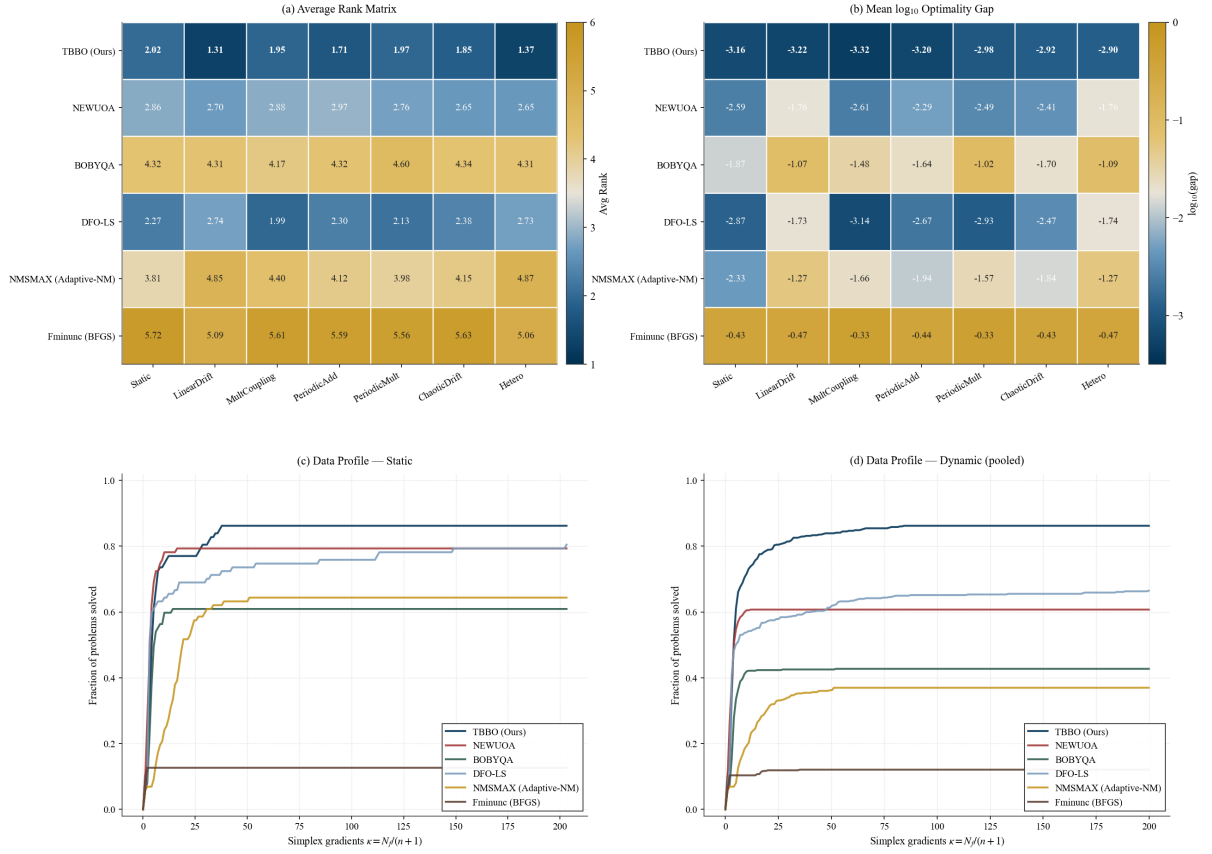


Figure 2: Moré–Wild benchmark at  $n = 6$ . (a) average rank matrix; (b) mean  $\log_{10}$  optimality gap; (c) static-environment data profile; (d) dynamic-environment data profile pooled over the six drift settings.

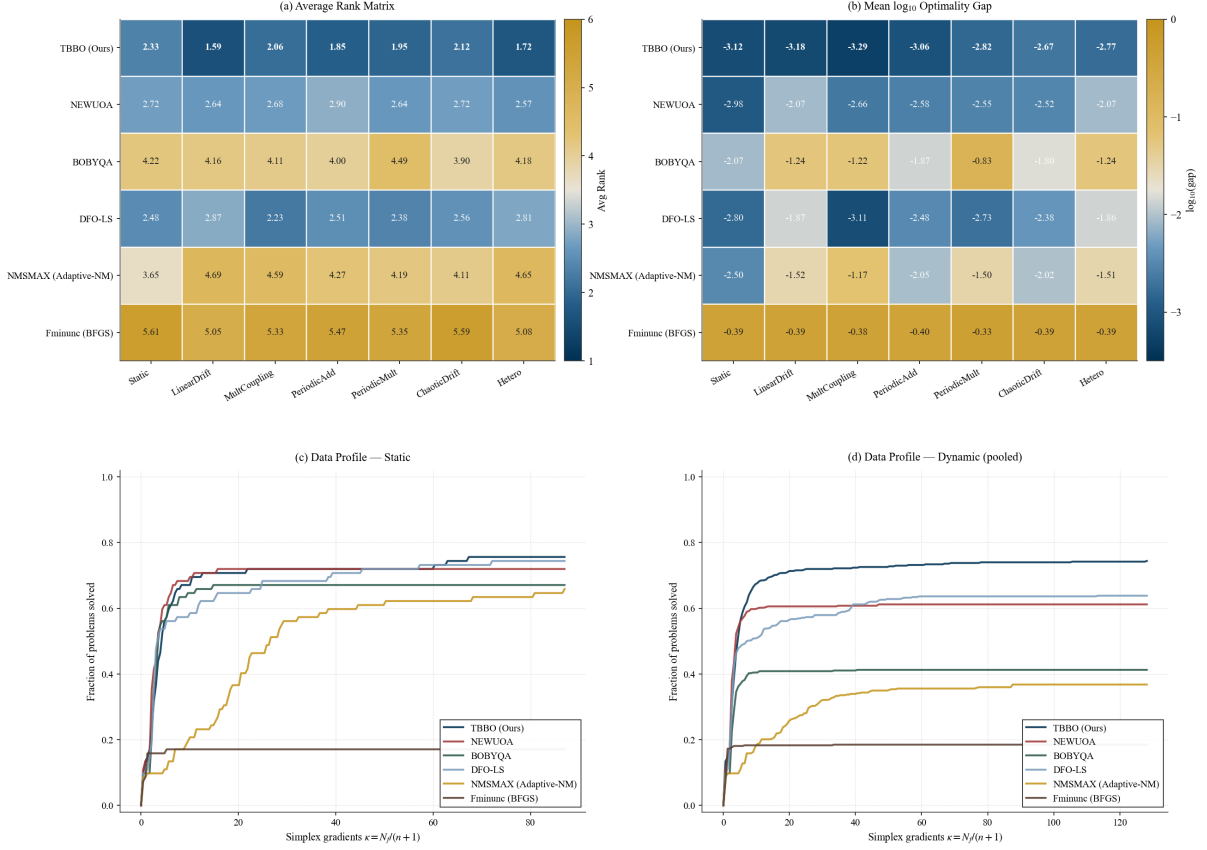


Figure 3: Moré–Wild benchmark at  $n = 10$ . Panels as in Figure 2.

axis (panel c). On LinearDrift the comparison is  $-3.22$  versus DFO-LS  $-1.73$  and NEWUOA  $-1.76$ , a margin of roughly 1.5 orders of magnitude. The pooled dynamic data profile (panel d) shows TOBYQA solving approximately 85% of problems within the full budget, against 67% for DFO-LS and 60% for NEWUOA. This separation is the empirical signature of Theorem 4.2: under additive linear drift, the augmented KKT system absorbs the temporal contamination of  $F_{\text{old}} - F_{\text{new}}$  through the recovered  $\gamma$ , so the trust-region update is driven by the static signal  $f(x_k) - f(x_k + s)$  that the baselines have to reconstruct from drifted observations. The margin narrows visibly on PeriodicMult ( $-2.98$  for TOBYQA versus  $-2.93$  for DFO-LS) and ChaoticDrift ( $-2.92$  versus  $-2.47$ ), consistent with the  $L_\mu \tau_{\max}^2$  bias predicted by Theorem 4.4 when the drift channel has nonzero second derivative.

At  $n = 10$  (Figure 3) the picture is quantitatively similar, with the absolute gaps reduced as the problem dimension grows. TOBYQA’s mean  $\log_{10}$  gap is  $-3.12$  on Static and  $-3.18$  on LinearDrift, against  $-2.80$  and  $-1.87$  respectively for DFO-LS; the ratio between the LinearDrift and Static gaps for TOBYQA is essentially unity, confirming that the drift machinery operates at the same precision as the underlying quadratic model. The dynamic data profile shows TOBYQA at approximately 74% solved against 64% for DFO-LS and 61% for BOBYQA. The relative position of the baselines shifts: BOBYQA overtakes DFO-LS on the Dynamic profile in this dimension, while DFO-LS remains stronger on the Static panel.

At  $n = 20$  (Figure 4) the dimension reaches the regime where quadratic-interpolation methods are known to lose efficiency relative to their low-dimensional behavior, since the interpolation set size  $m = 2n + 1$  becomes a more demanding fraction of the evaluation budget. TOBYQA retains the top rank on every environment but the margin compresses: the mean  $\log_{10}$  gap on LinearDrift is  $-2.84$  for TOBYQA against  $-1.74$  for DFO-LS, 1.1 orders of magnitude rather than the 1.5 observed at  $n = 6$ . The Dynamic data profile peaks around 0.69 for TOBYQA

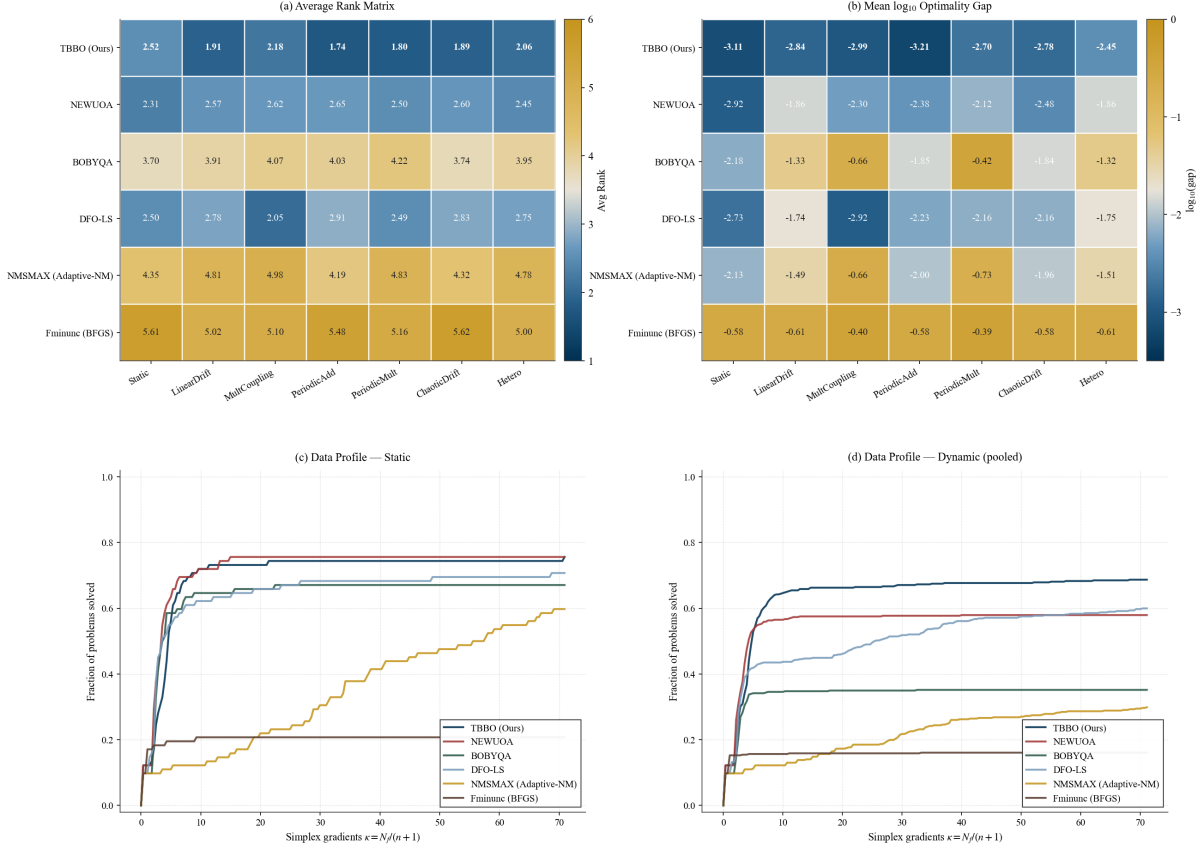


Figure 4: Moré–Wild benchmark at  $n = 20$ . Panels as in Figure 2.

at the budget ceiling, indicating that roughly thirty percent of problems at this dimension are not fully solved within 1500 evaluations by any of the methods tested. The behavior is not specific to TOBYQA—NEWUOA and DFO-LS show analogous compression—and reflects the dimensional fade that is a known characteristic of model-based DFO at this scale.

BFGS with finite-difference gradients solves fewer than 20% of instances at full budget on every dynamic environment, at every dimension, with mean  $\log_{10}$  gap above  $-0.5$  throughout. The gap between BFGS and the derivative-free cluster is the cost of finite-difference differentiation on a noisy time-varying landscape.

### 5.3 Component Ablation

To isolate the contribution of the two principal ingredients of TOBYQA—the augmented temporal column  $\theta$  together with the drift unknown  $\gamma$  (the *DriftModel*), and the diagonal affine-scaling transformation of Section 2.4 (the *AffineScale*)—we run a controlled ablation on the 29-function synthetic suite. Four variants are compared: the full algorithm, two single-component leave-one-out variants ( $-DriftModel$  and  $-AffineScale$ ), and the combined ablation ( $-Drift\&AS$ ) that disables both mechanisms simultaneously. Each (problem, environment, variant) is run with 3 seeds at  $\sigma = 0.1$  and dimensions  $n \in \{6, 10\}$ . We report the geometric-mean optimality gap with bootstrap 95% confidence intervals and the Wilcoxon signed-rank  $p$ -value against Full, paired at the problem level.

At  $n = 6$  (Figure 5), disabling the *DriftModel* raises the geometric-mean gap by a factor of 1.4–1.6 across every environment, with  $p < 0.001$  on Static, Dynamic, and Tidal and  $p < 0.01$  on Coupled. The Static column ( $+0.017$  gap units,  $p < 10^{-3}$ ) is worth noting: even on a no-drift problem, the augmented temporal column contributes significantly, indicating that the joint recovery of  $(c, g, G, \gamma)$  stabilizes the model fit beyond what removing the drift unknown

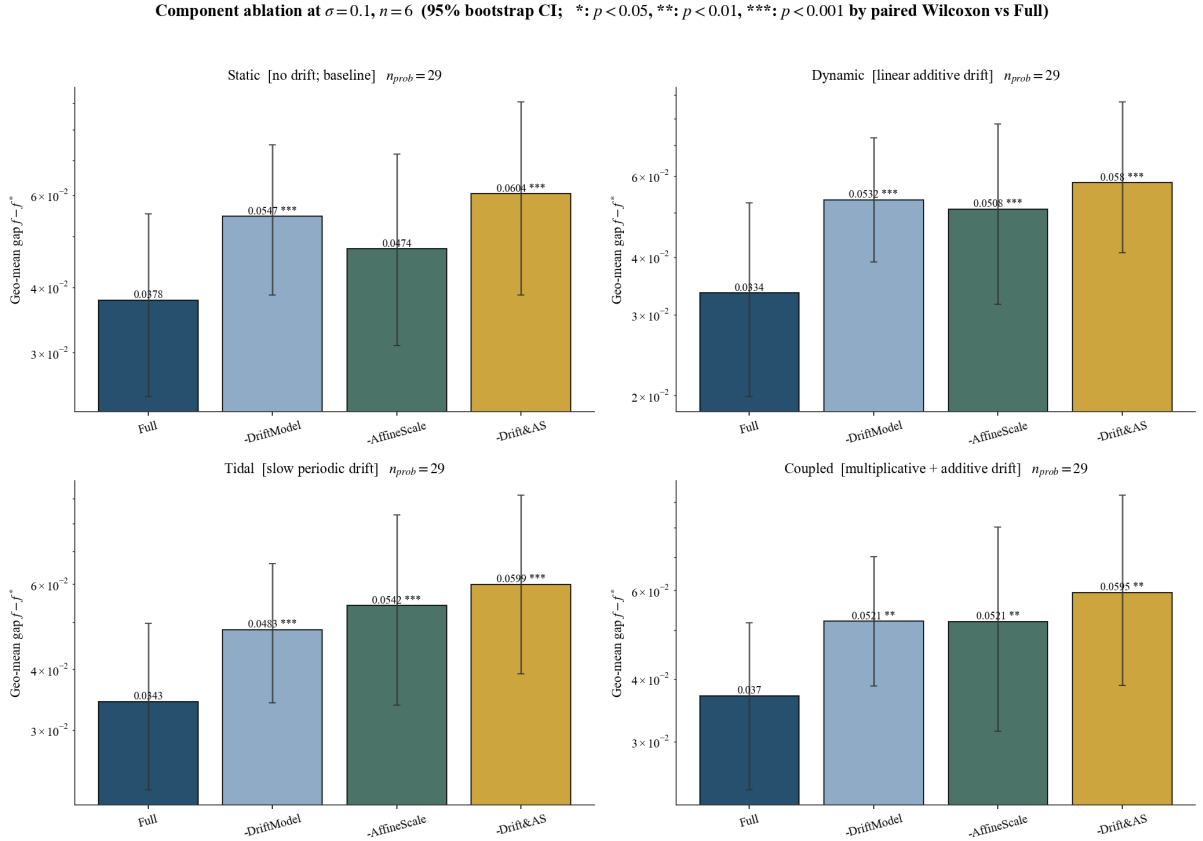


Figure 5: Component ablation at  $n = 6, \sigma = 0.1$ , 29 problems. Bars show the geometric-mean optimality gap with bootstrap 95% CI; significance markers report Wilcoxon signed-rank  $p$ -values paired against Full at the problem level (\* :  $p < 0.05$ , \*\* :  $p < 0.01$ , \*\*\* :  $p < 0.001$ ).

Component ablation at  $\sigma=0.1$ ,  $n=10$  (95% bootstrap CI; \*:  $p<0.05$ , \*\*:  $p<0.01$ , \*\*\*:  $p<0.001$  by paired Wilcoxon vs Full)

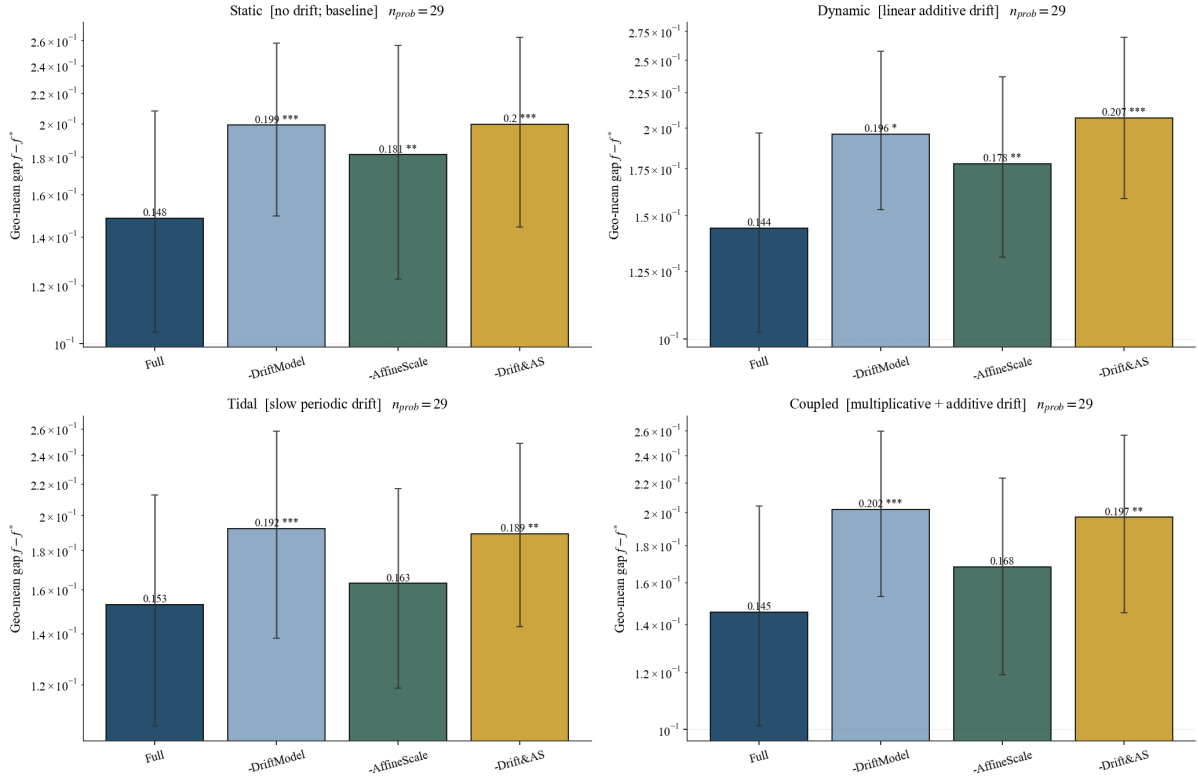


Figure 6: Component ablation at  $n=10$ , otherwise as Figure 5.

would suggest. On Dynamic and Coupled the gap rises by 60%, consistent with the prediction of Theorem 4.2: removing the drift unknown forces the algorithm to interpret the linear-in- $t$  component of  $F$  through the spatial unknowns alone, and the resulting model fit is correspondingly degraded.

The AffineScale contribution is smaller in aggregate—between +0.012 and +0.020 across environments at  $n=6$ , significant on Dynamic, Tidal, and Coupled but not on Static—but the per-group decomposition (Figure 7) reveals that AffineScale’s contribution is concentrated where the theory predicts. On the anisotropic-convex group (group B, 5 problems with strongly direction-dependent curvature), removing AffineScale raises the gap from  $\sim 0.04$  to  $\sim 0.10$ , a factor of 2–3, on every environment. On the isotropic-convex group (group A, 4 problems with near-spherical curvature) AffineScale has nearly no effect. This is the expected behavior of a diagonal curvature preconditioner: it delivers its full contribution on problems whose curvature is genuinely anisotropic, and idles on problems whose Hessian is already well-conditioned.

The combined ablation –Drift&AS produces a degradation approximately equal to the sum of the two single-component ablations: at  $n=6$ , 0.06 on Static (versus 0.055 from –DriftModel and 0.047 from –AffineScale, on a Full baseline of 0.038), and similarly on the dynamic environments. The absence of a strong interaction term indicates that the two mechanisms operate on orthogonal sources of difficulty: the DriftModel handles temporal contamination of the observations, AffineScale handles spatial anisotropy of the curvature, and neither needs the other to deliver its contribution. Results at  $n=10$  (Figure 6) follow the same pattern, with the AffineScale contribution becoming statistically significant in more cells—consistent with anisotropy effects becoming more prominent as  $n$  grows.

Ablation by problem group at  $\sigma=0.1, n=6$

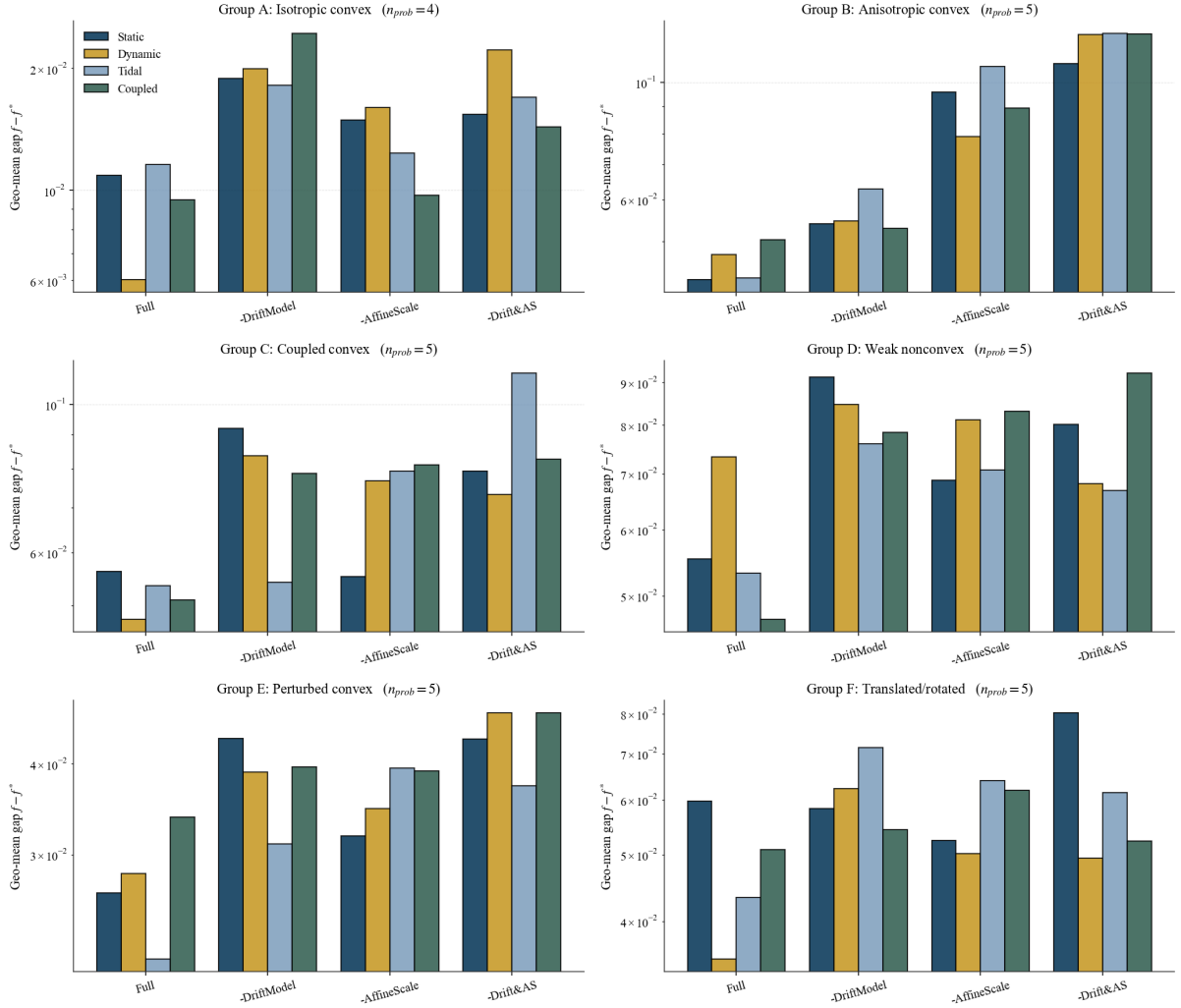


Figure 7: Component ablation broken down by problem group at  $n = 6$ . Each panel shows the geometric-mean gap of the four variants across the four environments (Static, Dynamic, Tidal, Coupled).

Sensitivity to  $\rho$  ( $n=6$ , 8 problems, error bars: 95% bootstrap CI)

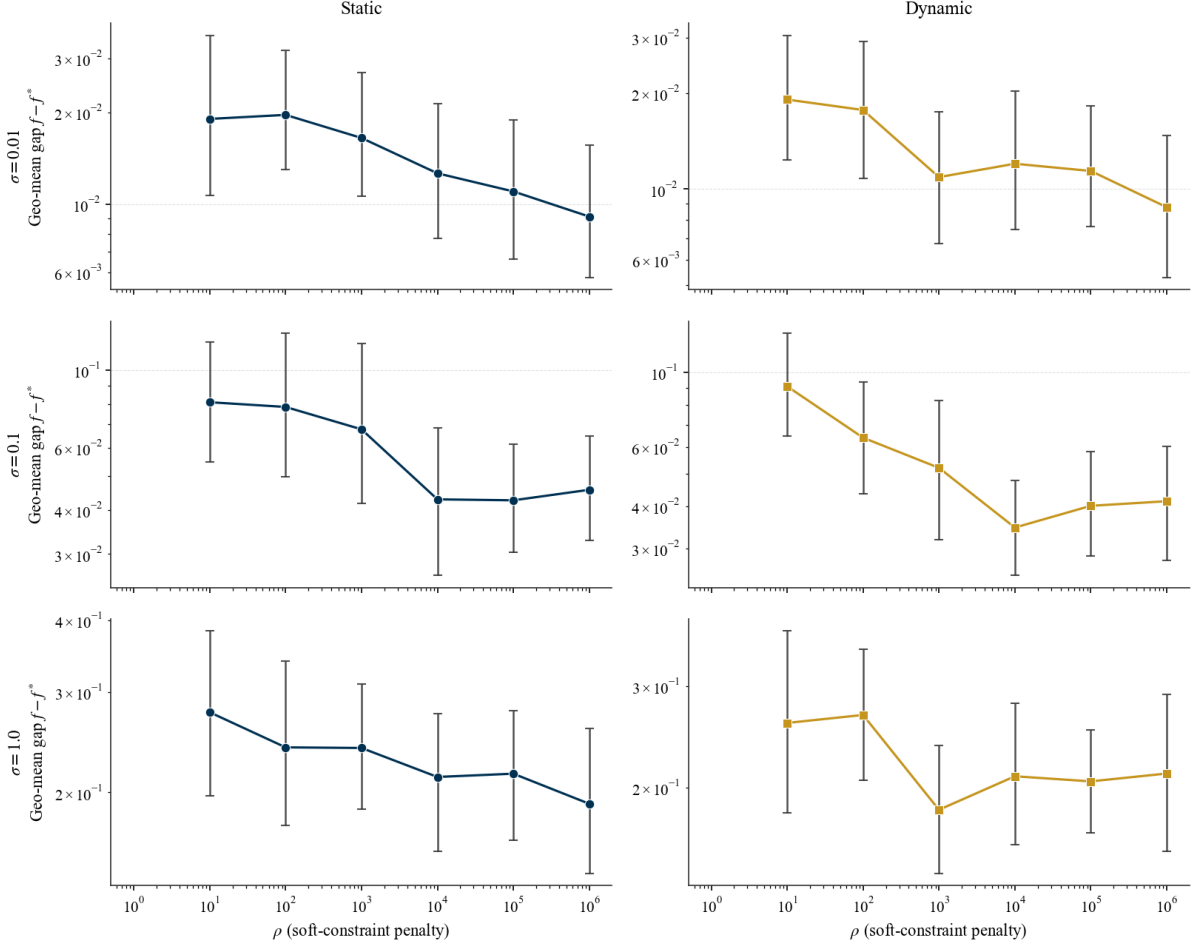


Figure 8: Sensitivity to the soft-constraint penalty  $\rho$ . Three rows correspond to  $\sigma \in \{0.01, 0.1, 1.0\}$ ; left and right columns to the Static and Dynamic environments. Error bars are 95% bootstrap CI.

#### 5.4 Hyperparameter Sensitivity

The algorithm exposes two user-facing parameters: the soft-constraint penalty  $\rho$  and the initial trust-region radius  $\Delta_0$ . We characterize the sensitivity of TOBYQA to each parameter by sweeping it across two orders of magnitude on a representative subset of the synthetic suite (group A, isotropic-convex, 4 problems), at three noise levels  $\sigma \in \{0.01, 0.1, 1.0\}$  and dimension  $n = 6$ , with 3 seeds.

The dependence on  $\rho$  (Figure 8) is monotonically decreasing across the entire range tested, on both Static and Dynamic environments at every noise level. The geometric-mean gap drops by roughly a factor of two between  $\rho = 10$  and  $\rho = 10^6$ , with most of the improvement concentrated below  $\rho = 10^4$  and a wide plateau above. The absence of an interior optimum is consistent with the hard-constraint limit of Proposition 2.9: at finite  $\rho$  the soft constraint trades a smaller Frobenius norm of the recovered Hessian against residual fit error, and as  $\rho$  grows the latter dominates, pulling the solution toward the exact interpolation regime. The plateau in  $[10^4, 10^6]$  indicates that the algorithm is robust to the choice of  $\rho$  within a wide range, and any value in this range serves as a reasonable default.

The dependence on  $\Delta_0$  (Figure 9) is noise-dependent. At low noise ( $\sigma = 0.01$ ) the gap is nearly flat across  $\Delta_0 \in [0.5, 20]$ . At moderate and high noise ( $\sigma = 0.1$  and  $\sigma = 1.0$ ) the optimum

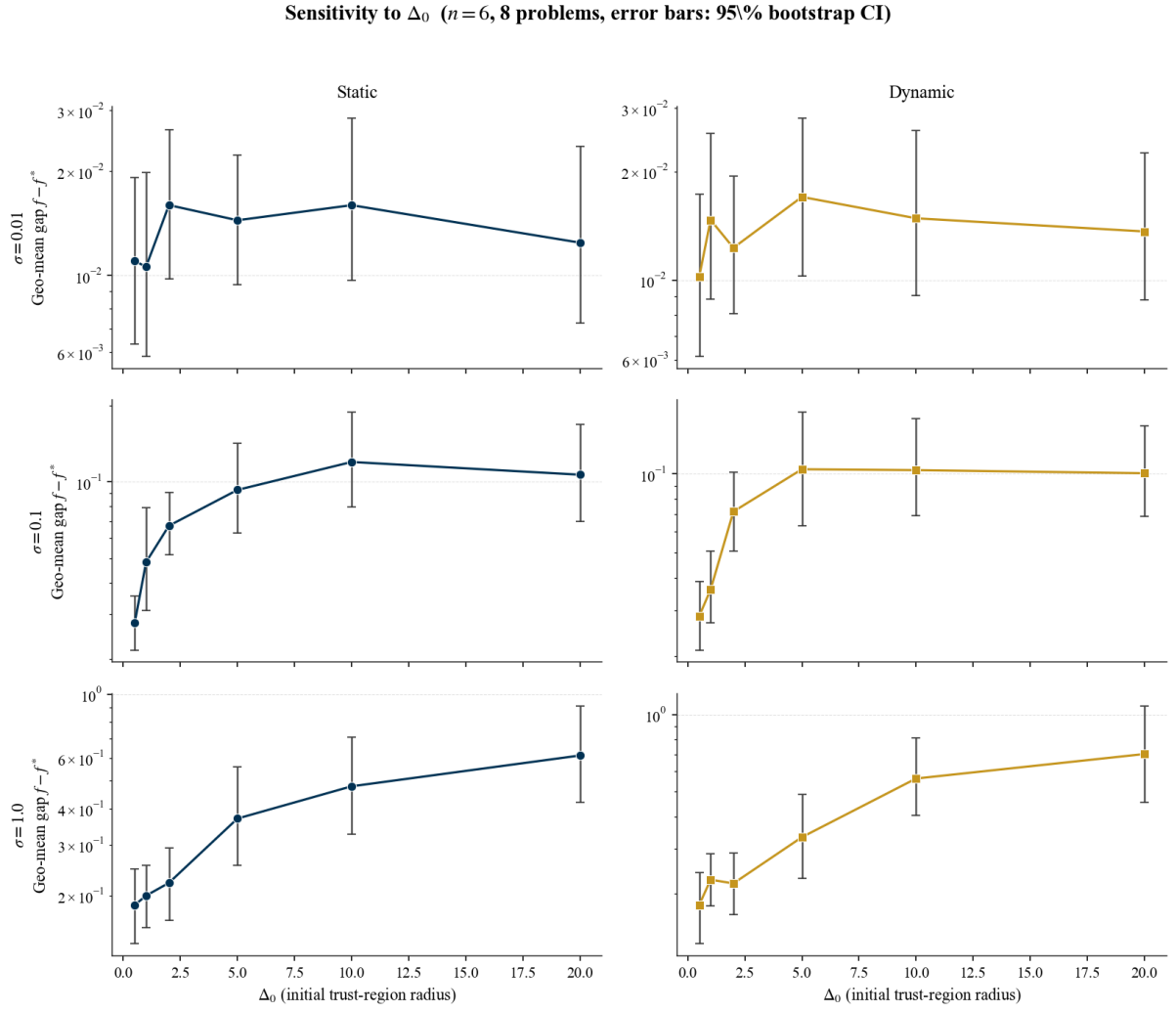


Figure 9: Sensitivity to the initial trust-region radius  $\Delta_0$ . Panels as in Figure 8.

lies at the small end of the range tested: the gap at  $\Delta_0 = 0.5$  is roughly half that at  $\Delta_0 = 20$ , and the monotone increase is sharp. The interpretation is that the trust-region machinery contracts  $\Delta_k$  as iterations proceed regardless of  $\Delta_0$ , but a large  $\Delta_0$  commits more of the early budget to exploration at radii where the noise-to-signal ratio is unfavorable. Choosing  $\Delta_0 = 1$  as the default is consistent with the small end of the sensitivity curve on the noise levels of practical interest.

## 6 Conclusion

We presented TOBYQA, a trust-region method for derivative-free optimization of noisy time-varying black-box objectives. The method extends the NEWUOA quadratic-interpolation framework by augmenting the interpolation system with a linear-in-time column that introduces a drift unknown  $\gamma$  alongside the spatial unknowns, replacing the hard interpolation constraints with a soft penalty that absorbs observation noise, and conditioning the trust-region subproblem in coordinates derived from the diagonal of the recovered Hessian. The three modifications are unified within a single augmented KKT system whose closed-form solution is the central object of the analysis.

The recovery operator  $\Pi$  associated with this system admits a simple algebraic structure: under the column-rank condition,  $\Pi\theta = (0, 0_n^\top, 1)^\top$  (Lemma 4.1). This identity drives the two recovery results of Section 4. Under additive linear drift  $F(x, t) = f(x) + \beta t$ , the recovered drift coefficient decomposes additively as  $\gamma = \gamma^{\text{stat}} + \beta$ , where  $\gamma^{\text{stat}}$  is the static-data bias that the same KKT system would produce in the absence of drift, and the recovered spatial gradient  $g$  is independent of the true rate  $\beta$  (Theorem 4.2). Under additive  $C^2$  drift  $\mu(t)$ , the recovery captures the instantaneous slope  $\mu'(t_k)$  with a curvature-of-drift bias of order  $L_\mu \tau_{\max}^2$  (Theorem 4.4). The exactness of the linear-drift recovery in the hard-constraint limit (Corollary 4.3) is the algebraic content of the drift-compensated ratio test.

The empirical evaluation supports these recovery results. On the Moré–Wild benchmark of 88 problems wrapped in seven temporal environments, TOBYQA is the top-ranked method across all three dimensions tested, with a margin of approximately 1.5 orders of magnitude in mean  $\log_{10}$  optimality gap over the strongest baseline on the additive-linear-drift environment and a narrowing margin on environments with nonlinear temporal structure. The component ablation on a 29-function synthetic suite identifies the augmented KKT system as the dominant contributor under drift, and the affine-scaling transformation as the dominant contributor on anisotropic curvature, with the two mechanisms operating on orthogonal sources of difficulty.

Several questions remain open. A tracking error bound for the iterate  $x_k$  relative to the time-varying minimizer requires a Lyapunov-style propagation argument that goes beyond the per-iteration recovery results established here. Global convergence of TOBYQA under the time-priority replacement rule is not addressed in this work: the classical NEWUOA convergence theory applies in the static regime under a  $\Lambda$ -poisedness maintenance rule, and extending it to the replacement rule actually used by the algorithm is left for future investigation. Finally, the linear-in-time temporal column captures only the local first derivative of the drift channel; enriching it with polynomial columns  $[\theta, \theta^{(2)}, \dots, \theta^{(p)}]$  would extend the exact-recovery regime to polynomial drift of order  $p$  at the cost of  $p - 1$  additional unknowns in the KKT system, and the trade-off between fidelity and conditioning of the enlarged system is a natural direction for further work.

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